MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLIER, DIGITAL, KEY CODE AGA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplier module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.72 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AGA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/268-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements specified in 3.10.1 through 3.10.3.

3.10.1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables II and VII satisfy or exceed these unique test patterns, no additional tests shall be added; otherwise, the required tests shall be added to table VII. "Stuck at H state" indicates a failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at H state" or a "stuck at L state", the vendor shall be required to submit to the procuring activity and the Standard Electronic Modules Program Quality Assurance Activity the proposed new set of test patterns for approval.

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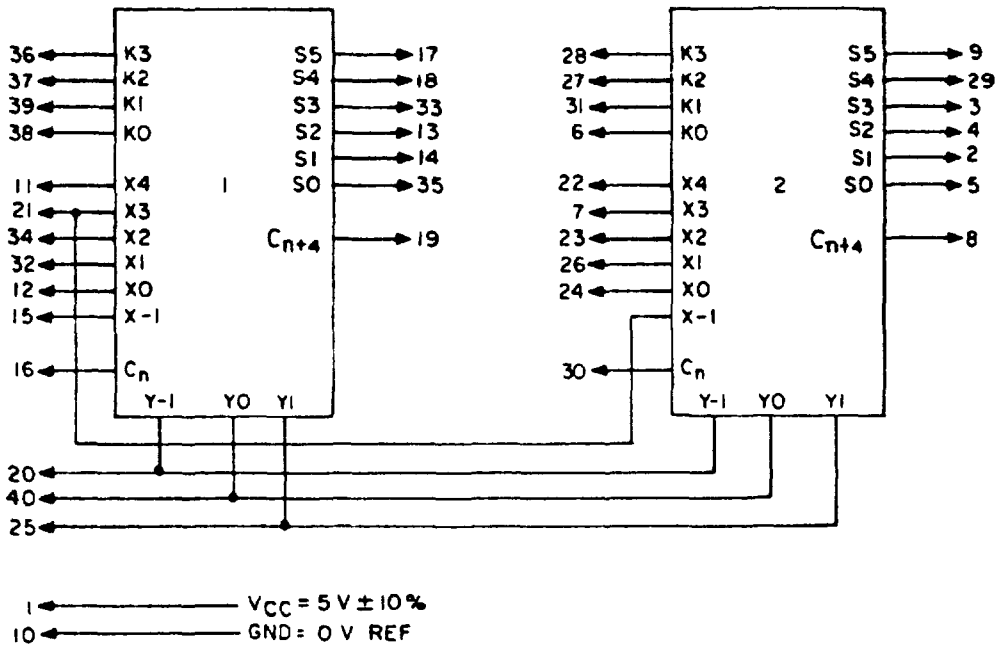


FIGURE 1. AGA module function diagram, multiplier.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A14	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1-B4,B6-B8, B11-B14,B17-B19,B21-B23	Input breakdown current	IV	S	0	2	0	2	0	2	0	2	mA
B5,B9,B10, B15,B16,B20	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
B24	Input breakdown current	IV	S	0	4	0	4	0	4	0	4	mA
C	Isolation	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D14	Output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
E1-E14	Output voltage, low level	VI	X	0	400	0	400	0	400	0	400	mV
F1-F21	Functional	VII	X	← Output vectors shall be as specified in table VII →								
G1-G6,G10-G19,G23-G28	Input current, low level	VIII	X	-0.8	-3.46	-0.8	-3.46	-0.8	-3.46	-0.8	-3.46	mA
G7-G9, G20-G22	Input current, low level	VIII	X	-0.4	-1.73	-0.4	-1.73	-0.4	-1.73	-0.4	-1.73	mA
G29	Input current, low level	VIII	X	-1.6	-6.92	-1.6	-6.92	-1.6	-6 92	-1.6	-6.92	mA
G30,G31	Input current, low level	VIII	X	-0.72	-3.12	-0.72	-3.12	-0.72	-3.12	-0.72	-3.12	mA
H1-H3, H11-H13	Input current, high level	IX	X	0	30	0	30	0	40	0	40	µA
H4-H10, H14-H21	Input current, high level	IX	X	0	60	0	60	0	80	0	80	µA
H22	Input current, high level	IX	X	0	120	0	120	0	160	0	160	µA
H23,H24	Input current, high level	IX	X	0	54	0	54	0	72	0	72	µA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J1-J14	Output short circuit current	X	X	-20	-85	-20	-85	-20	-85	-20	-85	mA
K1,K2	Power supply current (low-level and high-level)	XI	X	40	260	40	260	40	260	40	260	mA
L1-L14	Transition time (t_{TLH}), C_L = 75 pF	XII	S	4	30	4	30	4	35	4	35	ns
M1-M14	Transition time (t_{THL}), C_L = 75 pF	XIII	S	4	15	4	15	4	25	4	25	ns
N1,N2	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S	7	27	7	27	7	32	7	32	ns
N3-N10, N15-N42	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S,Q	7	29	7	29	7	34	7	34	ns
N11-N14	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S	8	32	8	32	8	38	7	38	ns
N43-N58	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S,Q	10	36	10	36	10	42	10	42	ns
N59-N68	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S,Q	11	37	11	37	11	43	11	43	ns
N69-N96	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S,Q	12	44	12	44	12	51	12	51	ns
N97-N120	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S,Q	13	44	13	44	13	56	13	56	ns
N121-N126	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S	19	59	19	59	19	74	19	74	ns
N127-N150	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S,Q	14	49	14	49	14	54	14	54	ns
N151-N162	Propagation delay time (t_{PHL}), C_L = 75 pF	XIV	S	16	51	16	51	16	63	16	63	ns

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See footnotes at end of table.

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*TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P1,P2	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S	6	26	6	26	6	31	6	31	ns
P3-P10, P23-P42	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S,Q	9	33	9	33	9	39	9	39	ns
P11-P14	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S	11	39	11	39	11	43	11	43	ns
P15-P22	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S,Q	6	24	6	24	6	29	6	29	ns
P43-P58	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S,Q	9	39	9	39	9	43	9	43	ns
P59-P68	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S,Q	10	38	10	38	10	42	10	42	ns
P69 P96	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S,Q	13	43	13	43	13	50	13	50	ns
P97-P120	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S,Q	16	51	16	51	16	63	16	63	ns
P121-P126	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S	14	49	14	49	14	56	14	56	ns
P127-P150	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S,Q	20	64	20	64	20	72	20	72	ns
P151 P162	Propagation delay time (t_{PLH}), C_L = 75 pF	XV	S	22	69	22	69	22	87	22	87	ns
Q1-Q24	Input clamp diode voltage	XVI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
R	Power supply filter capacitance	XVII	S	7.5	11.3	7.5	11.3	7.5	11.3	7.5	11.3	μF
S1-S20	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	pF

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S21-S23	Input capacitance	XVIII	Q	2	30	2	30	2	30	2	30	pF
S24	Input capacitance	XVIII	Q	2	60	2	60	2	60	2	60	pF
T	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	mV

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.8 V ±1%.
High level - Input voltage shall be 2 V ±1%.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows:

Low level - Output voltage shall be 0 V minimum, 0.4 V maximum.
High level - Output voltage shall be 2.4 V minimum, V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through R and T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through R and T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/268 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833572. Part number M28787/268-1 supersedes part numbers 2833572 and M28787/268.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation (all outputs at low level)		1,430 mW (outputs not loaded)
Supply voltage		5 V dc ±10 percent
Input voltage		
Low level		-0.5 V minimum, 0.8 V maximum
High level		2 V minimum, V_{CC} maximum
Input current		
X0-X2, K0-K3, (1X3, 2X-1)		
Low level	(V_{IL} = 0 V)	-3.46 mA maximum
	(V_{IL} = 0.4 V)	-3.2 mA maximum
High level	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	2 mA maximum
X4, 1X-1, C_n, 2X3		
Low level	(V_{IL} = 0 V)	-1.73 mA maximum
	(V_{IL} = 0.4 V)	-1.6 mA maximum
High level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
Y0		
Low level	(V_{IL} = 0 V)	-6.92 mA maximum
	(V_{IL} = 0.4 V)	-6.4 mA maximum
High level	(V_{IH} = 2.4 V)	160 µA maximum
	(V_{IH} = 5.5 V)	4 mA maximum
Y1, Y-1		
Low level	(V_{IL} = 0 V)	-3.12 mA maximum
	(V_{IL} = 0.4 V)	-2.88 mA maximum
High level	(V_{IH} = 2.4 V)	72 µA maximum
	(V_{IH} = 5.5 V)	2 mA maximum
Output voltage		
Low level	(I_{OL} = 9.6 mA)	0.4 V maximum
High level	(I_{OH} = -480 µA)	2.4 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.4 V)	9.6 mA
High level	(V_{OH} = 2.4 V)	-480 µA

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* 6.5 Module function. This module contains two 2 x 4 bit 2's complement multiplier circuits which are intercornected to form one 2 x 8 bit multiplier circuit. Each multiplier circuit performs 2's complement multiplication in positive logic. Carry in (C_n) and carry out (C_{n+4}) are required to provide an iterative array capability. The circuit assumes that the most significant digit in a word carries a negative weight which allows use of the circuits in arrays where the multiplicand and multiplier have different word lengths. The operations performed on the "X" and "K" inputs are determined by the logic states of the multiplier inputs (Y1, Y0, Y-1) as shown in table II.

6.5.1 Parallel outputs. Paralleling outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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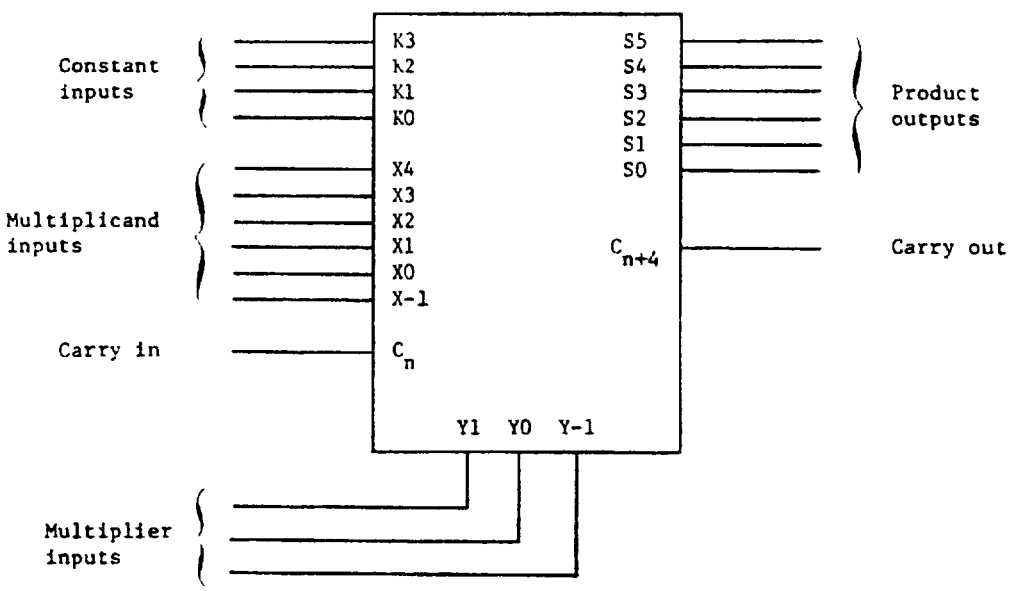
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TABLE II. Input to output relationship. 1/ 2/

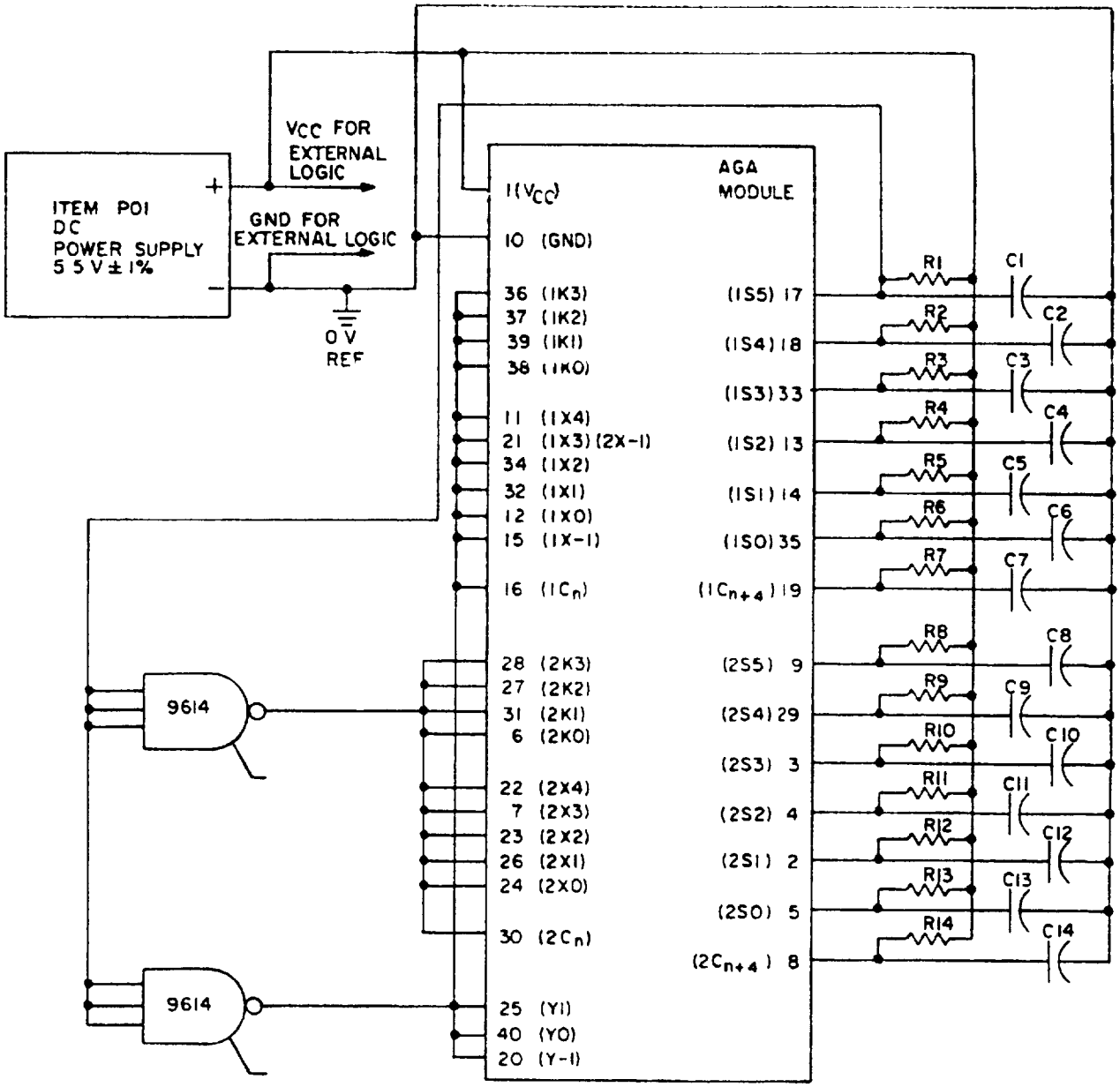
Y inputs			Outputs
Y-1	Y0	Y1	Operation performed
L	L	L	K + 0
H	L	L	K + X
L	H	L	K + Y
H	H	L	K + 2X
L	L	H	K - 2X
H	L	H	K - X
L	H	H	K - X
H	H	H	K - 0

1/ L = Low level.

2/ H = High level.



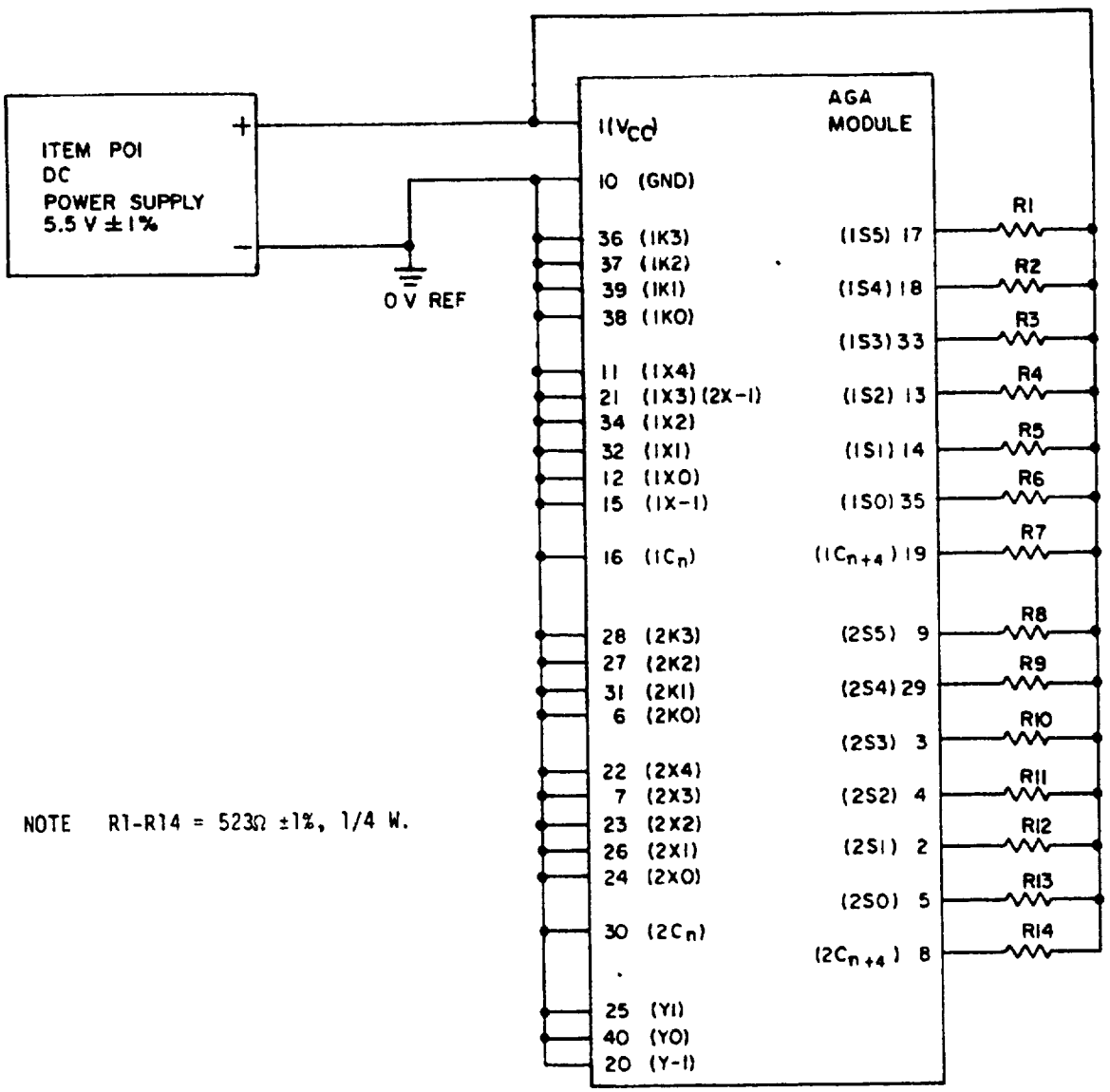
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NOTES
1 R1 = 680Ω ±1%, 1/4 W.
2 R2-R14 = 523Ω ±1%, 1/4 W
3 C1-C14 = 80 pF ±20%.
4 9614 = Dual differential line driver

FIGURE 2 Operating temperature and life test circuit

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NOTE R1-R14 = 523Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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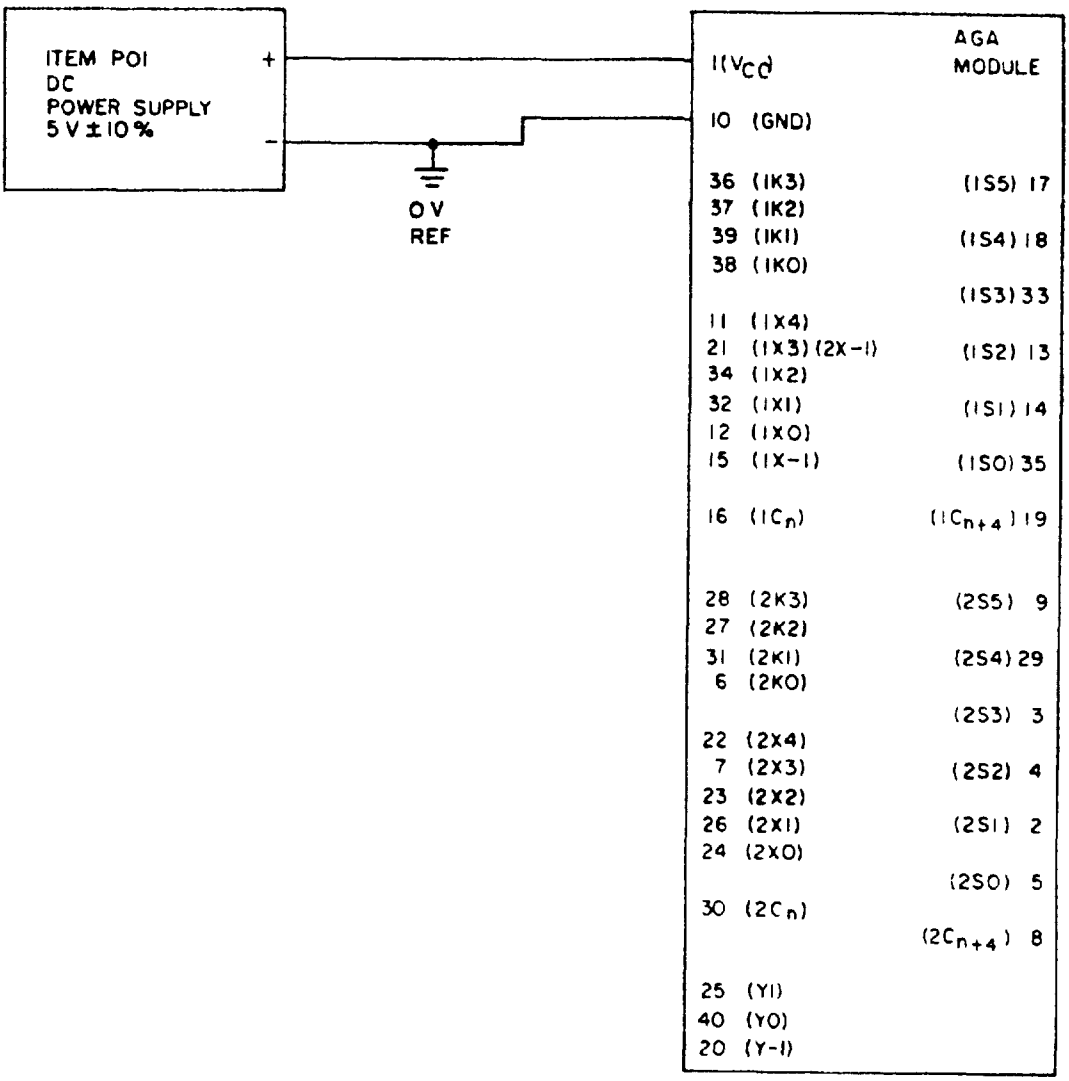


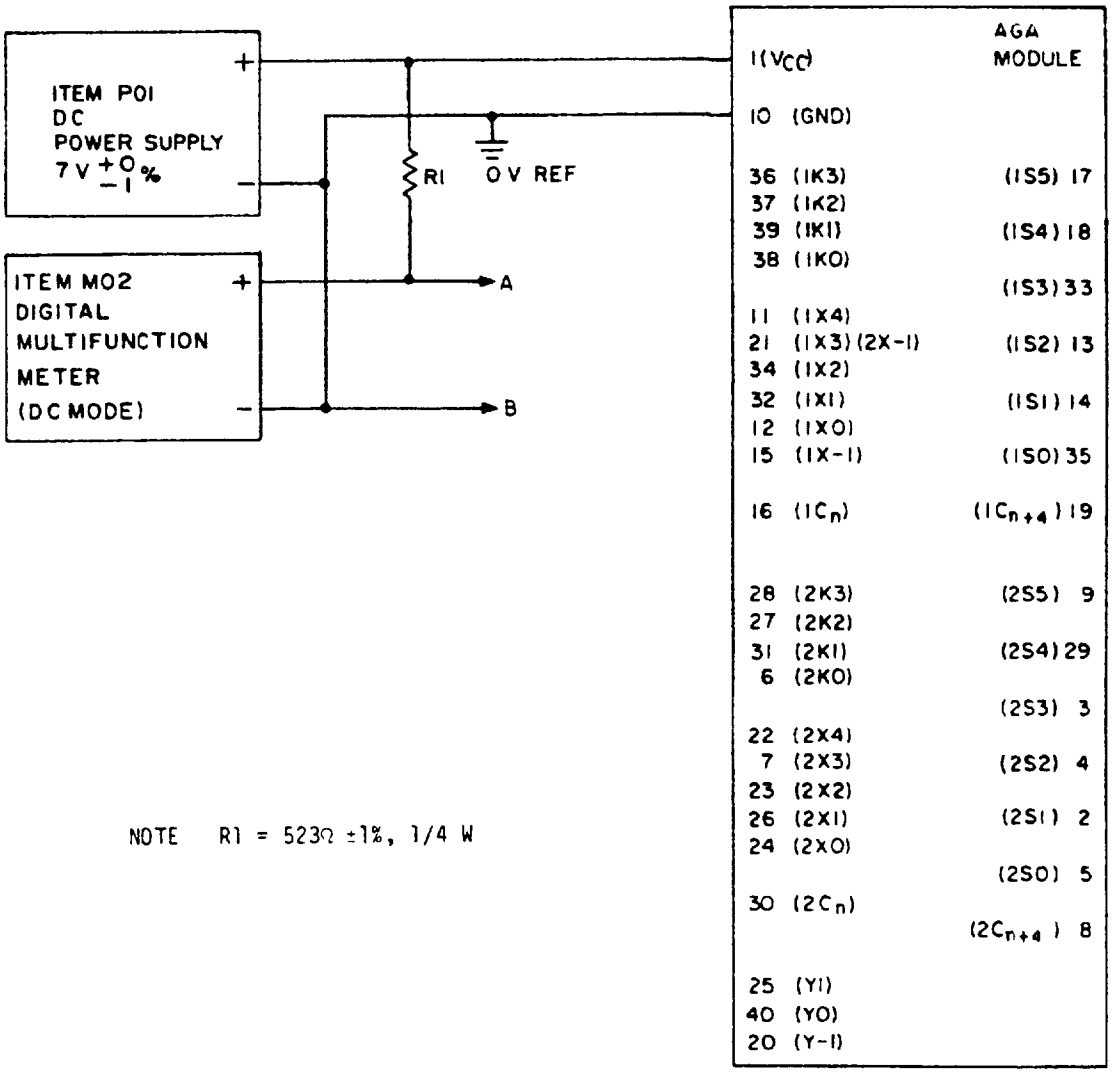
FIGURE 4 Vibration test circuit

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
A1	1S5	V_{OUT} (low loaded)	17	11,12,15,16,20,21, 25,32,34,36,37,38, 39,40	10	800	mV	5
A2	1S4	"	18	"	"	"	"	"
A3	1S3	"	33	"	"	"	"	"
A4	1S2	"	13	"	"	"	"	"
A5	1S1	"	14	"	"	"	"	"
A6	1S0	"	35	"	"	"	"	"
A7	$1C_{n+4}$	"	19	"	"	"	"	"
A8	2S5	"	9	6,7,20,21,22,23,24, 25,26,27,28,30,31, 40	"	"	"	"
A9	2S4	"	29	"	"	"	"	"
A10	2S3	"	3	"	"	"	"	"
A11	2S2	"	4	"	"	"	"	"
A12	2S1	"	2	"	"	"	"	"
A13	2S0	"	5	"	"	"	"	"
A14	$2C_{n+4}$	"	8	"	"	"	"	"

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NOTE R1 = 523Ω ±1%, 1/4 W

* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1K3 Input breakdown current	36	20	0	2	mA	F
B2	1K2 "	37	"	"	"	"	"
B3	1K1 "	39	"	"	"	"	"
B4	1K0 "	38	"	"	"	"	"
B5	1X4 "	11	Open	"	1	"	"
B6	1X2 "	34	"	"	2	"	"
B7	1X1 "	32	"	"	"	"	"
B8	1X0 "	12	"	"	"	"	"
B9	1X-1 "	15	"	"	1	"	"
B10	$1C_n$ "	16	"	"	1	"	"
B11	2K3 "	28	20	"	2	"	"
B12	2K2 "	27	"	"	"	"	"
B13	2K1 "	31	"	"	"	"	"
B14	2K0 "	6	"	"	"	"	"
B15	2X4 "	22	Open	"	1	"	"
B16	2X3 "	7	"	"	1	"	"
B17	2X2 "	23	"	"	2	"	"
B18	2X1 "	26	"	"	"	"	"
B19	2X0 "	24	"	"	"	"	"
B20	$2C_n$ "	30	"	"	1	"	"
B21	1X3,2X-1 "	21	"	"	2	"	"

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B22	Y1	Input breakdown current	25	Open	0	2	mA	6
B23	Y-1	"	20	"	"	2	"	"
B24	Y0	"	40	"	"	4	"	"

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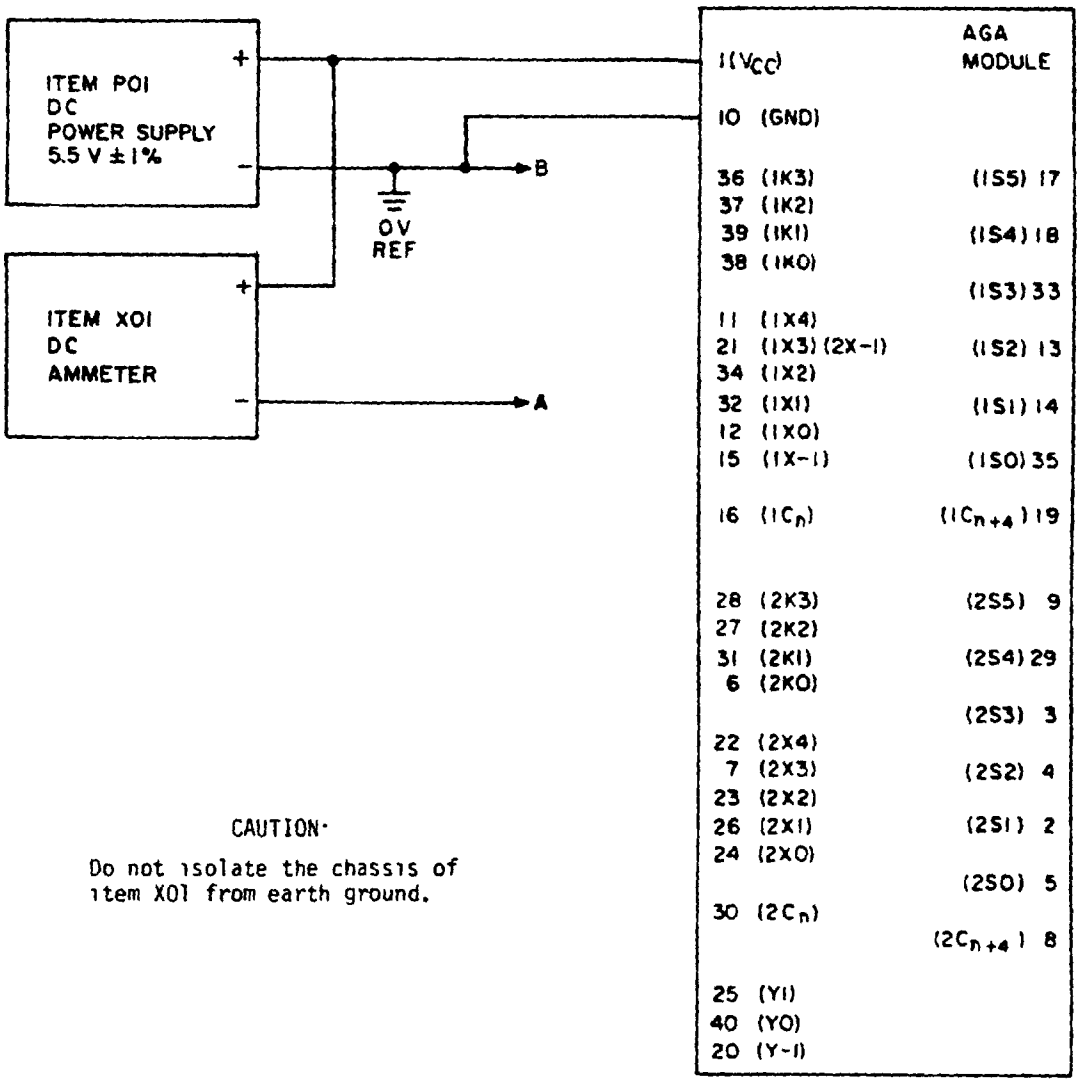


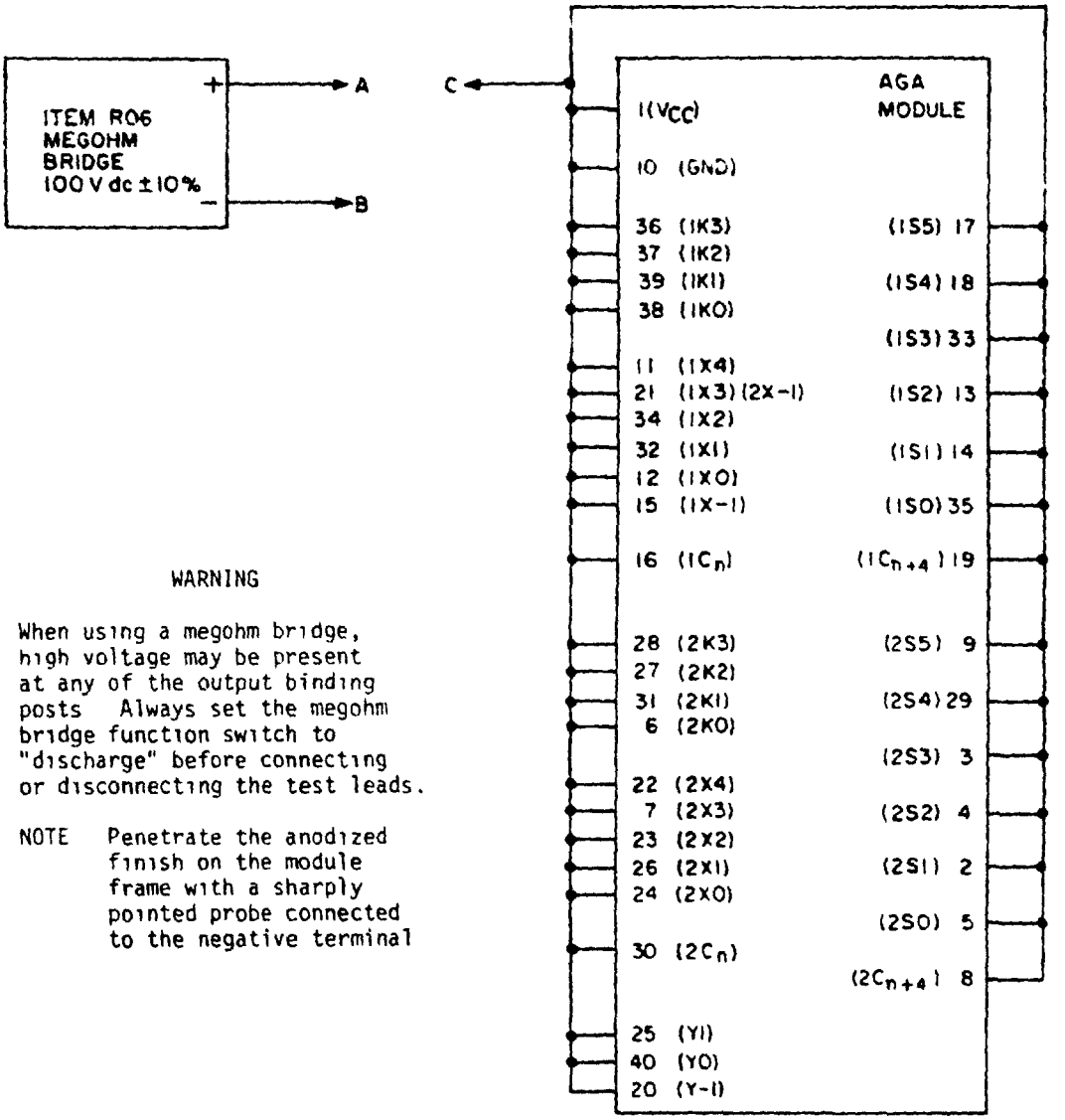
FIGURE 6 Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	C	Frame	10	∞	MΩ	7

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 7 Isolation test circuit for table V.

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*TABLE VI Output voltage (high-level and low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	1S5	Output voltage, high level	36,37,39,38, 11,21,34,32, 12,15,16,20, 40,25	Open	17	2.4	4.5	V	8
D2	1S4	"	"	"	18	"	"	"	"
D3	1S3	"	"	"	33	"	"	"	"
D4	1S2	"	"	"	13	"	"	"	"
D5	1S1	"	"	"	14	"	"	"	"
D6	1S0	"	"	"	35	"	"	"	"
D7	$1C_{n+4}$	"	"	"	19	"	"	"	"
D8	2S5	"	28,27,31,6, 22,7,23,26, 24,21,30,20, 40,25	"	9	"	"	"	"
D9	2S4	"	"	"	29	"	"	"	"
D10	2S3	"	"	"	3	"	"	"	"
D11	2S2	"	"	"	4	"	"	"	"
D12	2S1	"	"	"	2	"	"	"	"
D13	2S0	"	"	"	5	"	"	"	"
D14	$2C_{n+4}$	"	"	"	8	"	"	"	"
E1	1S5	Output voltage, low level	Open	36,37,39,38, 11,21,34,32, 12,15,16,20, 40,25	17	0	400	mV	"
E2	1S4	"	"	"	18	"	"	"	"
E3	1S3	"	"	"	33	"	"	"	"
E4	1S2	"	"	"	13	"	"	"	"

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*TABLE VI. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E5	1S1	Output voltage, low level	Open	36,37,39,38, 11,21,34,32, 12,15,16,20, 40,25	14	0	400	mV	8
E6	1S0	"	"	"	35	"	"	"	"
E7	$1C_{n+4}$	"	"	"	19	"	"	"	"
E8	2S5	"	Open	28,27,31,6, 22,7,23,26, 24,21,30,20, 40,25	9	"	"	"	"
E9	2S4	"	"	"	29	"	"	"	"
E10	2S3	"	"	"	3	"	"	"	"
E11	2S2	"	"	"	4	"	"	"	"
E12	2S1	"	"	"	2	"	"	"	"
E13	2S0	"	"	"	5	"	"	"	"
E14	$2C_{n+4}$	"	"	"	8	"	"	"	"

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TABLE VII. Functional test.

Test condition	Parameter	Circuit	Connect			Output logic states							Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	S0	S1	S2	S3	S4	S5	C_n+4	Min	Max		
						35	14	13	33	18	17	19				
						5	2	4	3	29	9	8				
F1	Functional	1 and 2	6,11,12,22,23,24, 25,26,27,28,31,32, 34,36,37,38,39	7,15,16,20, 21,30,40	2/ 3/	L	L	L	L	L	L	H		1/		8
F2	"	2	6,21,22,23,25,26, 27,28,31	7,20,24,30, 40	"	H	L	L	L	L	L	H		"		"
F3	"	1	11,15,25,32,34,36, 37,38,39	12,16,20, 21,40	"	H	L	L	L	L	L	H		"		"
F4	"	2	6,21,22,24,25,26, 27,28,31	7,20,23,30, 40	"	H	H	H	L	L	L	H		"		"
F5	"	1	11,12,15,25,32,36, 37,38,39	16,20,21, 34,40	"	H	H	H	L	L	L	H		"		"
F6	"	1 and 2	7,11,12,15,16,21, 22,23,24,25,26,27, 28,30,31,32,34,36, 37,39	6,20,38,40	"	H	H	H	H	H	H	L		"		"
F7	"	1 and 2	7,11,15,16,20,21, 22,23,25,26,27,28, 30,31,32,34,36,37, 39	6,12,24,38, 40	"	L	L	L	L	L	L	H		"		"
F8	"	1 and 2	7,11,12,15,16,20, 21,22,23,24,25,27, 28,30,31,34,36,37, 39	6,26,32,38, 40	"	H	L	L	L	L	L	H		"		"
F9	"	1 and 2	7,11,12,15,16,20, 21,22,24,25,26,27, 28,30,32,36,37	6,23,31,34, 38,39,40	"	H	L	L	L	L	L	H		"		"
F10	"	2	20,21,22,23,24,25, 26,28,30	6,7,27,31, 40	"	H	L	L	L	H	H	H		"		"
F11	"	1	11,12,15,16,20,25, 32,34,36	21,37,38, 39,40	"	H	L	L	L	H	H	H		"		"

See footnotes at end of table.

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TABLE VII. Functional test - Continued.

Test condition	Parameter	Circuit	Connect			Output logic states							Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	S0	S1	S2	S3	S4	S5	C_{n+4}	Min	Max		
						35	14	13	33	18	17	19				
						5	2	4	3	29	9	8				
F12	Functional	1 and 2	Open	6,7,11,12,15,16,20,21,22,23,24,25,26,27,28,30,31,32,34,36,37,38,39,40	2/ 3/	L	L	L	L	L	L	L		1/		8
F13	"	1 and 2	25	6,7,11,12,15,16,20,21,22,23,24,26,27,28,30,31,32,34,36,37,38,39,40	"	H	H	H	H	H	H	L		"		"
F14	"	2	6,21,22,23,24,25,27,28,31	7,20,26,30,40	"	H	H	L	L	L	L	H		"		"
F15	"	1	11,12,15,25,34,36,37,38,39	16,20,21,32,40	"	H	H	L	L	L	L	h		"		"
F16	"	1 and 2	6,7,11,12,15,16,21,22,24,25,26,27,28,30,31,32,36,37,38,39,40	20,23,34	"	L	L	H	L	L	L	H		"		"
F17	"	1 and 2	6,7,11,12,15,16,20,21,22,23,24,26,27,28,30,31,32,34,36,37,38,39,40	25	"	H	H	H	H	H	H	H		"		"
F18	"	1 and 2	6,7,11,12,15,16,21,22,23,24,26,27,30,31,32,34,37,38,39	20,25,28,36,40	"	L	L	L	H	L	L	L		"		"
F19	"	1 and 2	6,7,11,12,15,16,21,22,23,24,26,28,30,31,32,34,36,38,39	20,25,27,37,40	"	L	L	H	H	H	H	L		"		"

See footnotes at end of table.

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TABLE VII. Functional test - Continued

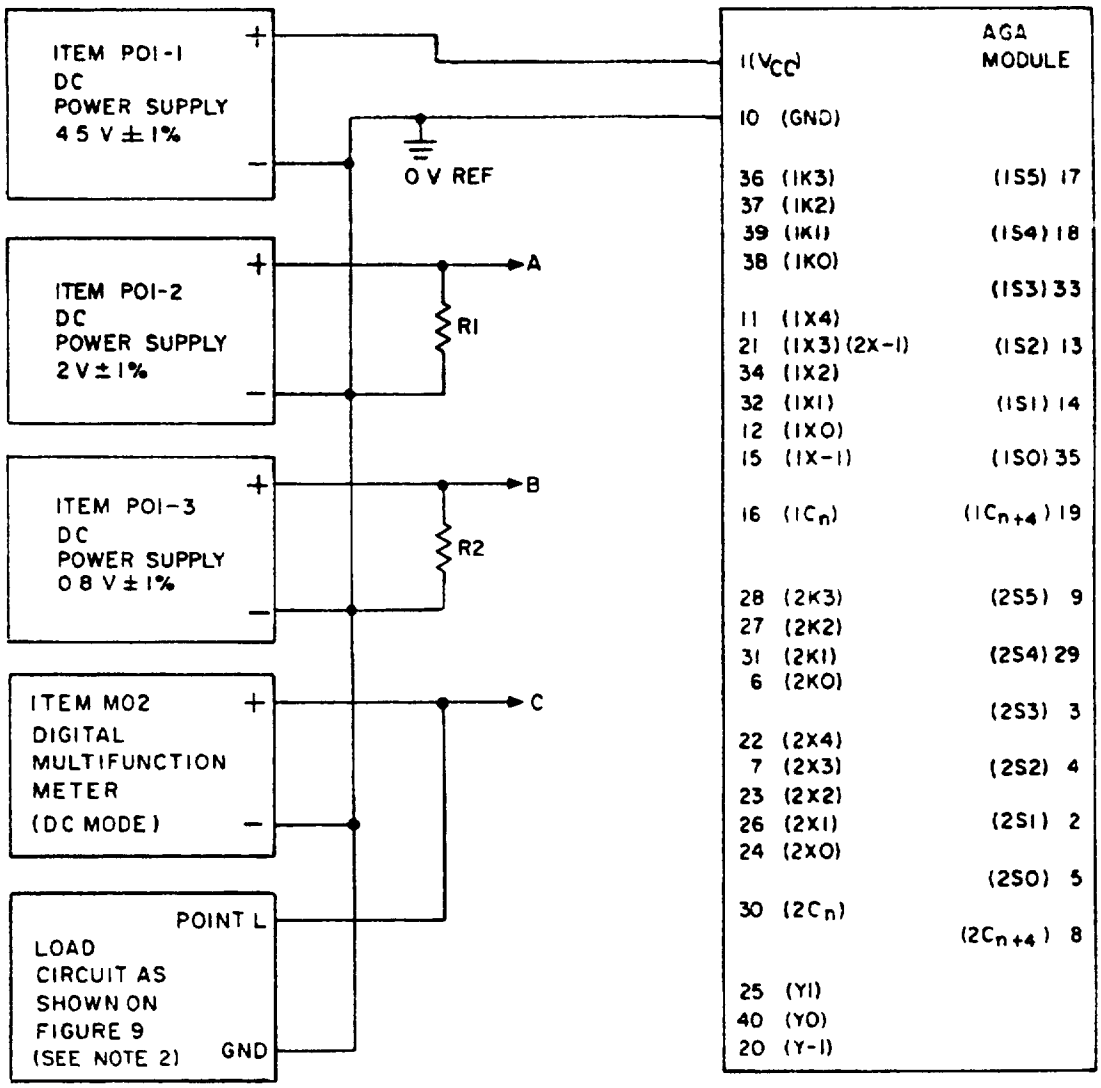
Test condition	Parameter	Circuit	Connect			Output logic states							Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	S0	S1	S2	S3	S4	S5	C_{n+4}	Min	Max		
						35	14	13	33	18	17	19				
						5	2	4	3	29	9	8				
F20	Functional	1 and 2	6,7,11,12,15,16, 21,22,23,24,26, 27,28,30,32,34, 36,37,38	20,25,31,39, 40	2/ 3/	L	H	H	H	H	H	L		1/		8
F21	Functional	1 and 2	7,11,15,16,20, 21,22,23,26,27, 30,31,32,34,37, 39	6,12,24,25, 28,36,38,40	2/ 3/	H	L	H	L	L	L	H		1/		8

1/ L = Low level = 0.0 to 0.4 V.
H = High level = 2.4 to 4.5 V.

2/ For test conditions F1-F21 verification of the output logic states in footnote 1 is obtained by connecting item M02, point C, to each output in succession after input conditions have been accomplished.

3/ S0, S1, S2, S3, S4, S5, and C_{n+4} output logic states are measured at pins 35, 14, 13, 33, 18, 17, and 19, respectively, for circuit 1 and pins 5, 2, 4, 3, 29, 9, and 8, respectively, for circuit 2.

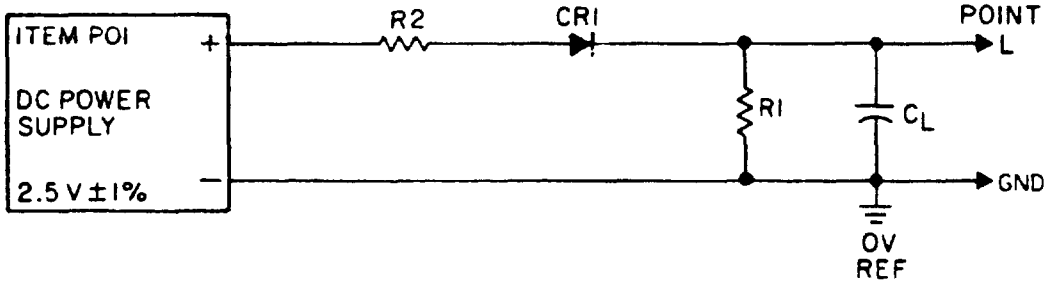
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. The load circuit as shown on figure 9 is not required for test conditions F1-F21.

* FIGURE 8. Output voltage (high-level and low-level) and functional test circuit for tables VI and VII.

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NOTES.
1. CR1 = 1N4150, or equivalent.
2. R1 = 5 kΩ ±1%, 1/4 W.
3. R2 = Select value for current flow of 9.6 mA ±1% out of point L when point L is held at 0.4 V.
4. C_L = 75 ±5 pF (including probe and parasitic capacitance).
5. C_L is not required for dc electrical tests D1-D14 and E1-E14.
6. The maximum lead length from point L to the appropriate module pin shall be 2 inches (50.8 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

* FIGURE 9. AGA logic load circuit

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1X2 Input current, low level	Open	34	25	-0.8	-3.46	mA	10
G2	1X2 "	"	34	20	"	"	"	"
G3	1X1 "	"	32	25	"	"	"	"
G4	1X1 "	"	32	20	"	"	"	"
G5	1XO "	"	12	25	"	"	"	"
G6	1XO "	"	12	20	"	"	"	"
G7	1X4 "	"	11	25	-0.4	-1.73	"	"
G8	1X-1 "	"	15	25	"	"	"	"
G9	$1C_n$ "	"	16	Open	"	"	"	"
G10	1KO "	"	38	"	-0.8	-3.46	"	"
G11	1K1 "	"	39	"	"	"	"	"
G12	1K2 "	"	37	"	"	"	"	"
G13	1K3 "	"	36	"	"	"	"	"
G14	2X2 "	"	23	25	"	"	"	"
G15	2X2 "	"	23	20	"	"	"	"
G16	2X1 "	"	26	25	"	"	"	"
G17	2X1 "	"	26	20	"	"	"	"
G18	2XO "	"	24	25	"	"	"	"
G19	2XO "	"	24	20	"	"	"	"
G20	2X4 "	"	22	25	-0.4	-1.73	"	"
G21	2X3 "	"	7	20	"	"	"	"
G22	$2C_n$ "	"	30	Open	"	"	"	"

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G23	2KO	Input current, low level	Open	6	Open	-0.8	-3.46	mA	10
G24	2K1	"	"	31	"	"	"	"	"
G25	2K2	"	"	27	"	"	"	"	"
G26	2K3	"	"	28	"	"	"	"	"
G27	1X3, 2X-1	"	"	21	25	"	"	"	"
G28	1X3, 2X-1	"	"	21	20	"	"	"	"
G29	YO	"	"	40	25	-1.6	-6.92	"	"
G30	Y1	"	"	25	Open	-0.72	-3.12	"	"
G31	Y-1	"	"	20	Open	-0.72	-3.12	"	"

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1X4 Input current, high level	11	Open	Open	0	30	µA	10
H2	1X-1 "	15	"	"	"	"	"	"
H3	$1C_n$ "	16	"	"	"	"	"	"
H4	1X2 "	34	"	"	"	60	"	"
H5	1X1 "	32	"	"	"	"	"	"
H6	1X0 "	12	"	"	"	"	"	"
H7	1K0 "	38	"	20	"	"	"	"
H8	1K1 "	39	"	"	"	"	"	"
H9	1K2 "	37	"	"	"	"	"	"
H10	1K3 "	36	"	"	"	"	"	"
H11	2X4 "	22	"	Open	"	30	"	"
H12	2X3 "	7	"	"	"	"	"	"
H13	$2C_n$ "	30	"	"	"	"	"	"
H14	2X2 "	23	"	"	"	60	"	"
H15	2X1 "	26	"	"	"	"	"	"
H16	2X0 "	24	"	"	"	"	"	"
H17	2K0 "	6	"	20	"	"	"	"
H18	2K1 "	31	"	"	"	"	"	"
H19	2K2 "	27	"	"	"	"	"	"
H20	2K3 "	28	"	"	"	"	"	"
H21	1X3, 2X-1 "	21	"	Open	"	"	"	"

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TABLE IX. Input current (high-level) test - Continued.

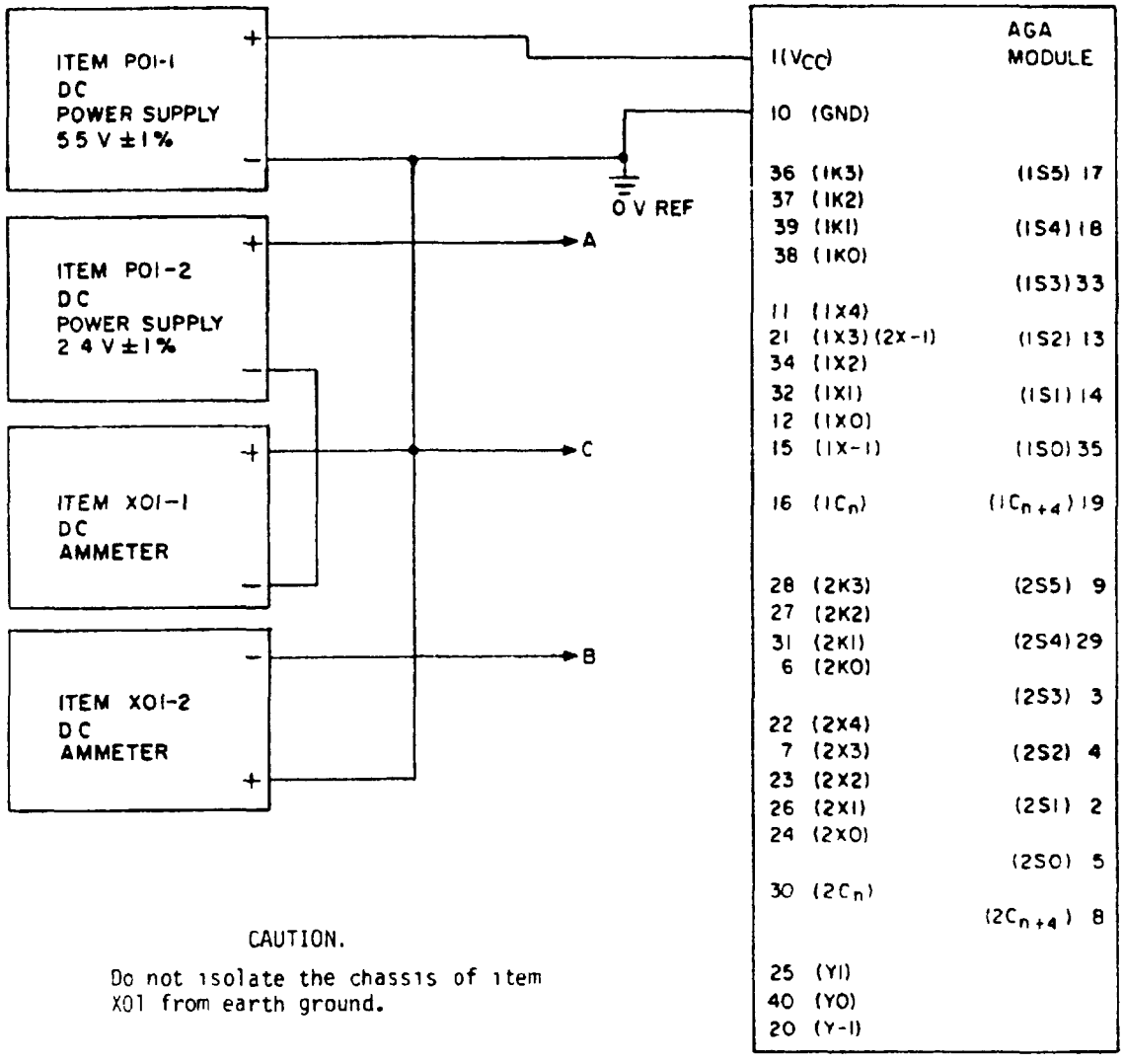
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H22	Y0	Input current, high level	40	Open	Open	0	120	µA	10
H23	Y1	"	25	"	"	"	54	"	"
H24	Y-1	"	20	"	"	"	54	"	"

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1S5 Output short circuit current	Open	17	Open	-20	-85	mA	10
J2	1S4 "	"	18	"	"	"	"	"
J3	1S3 "	"	33	"	"	"	"	"
J4	1S2 "	"	13	"	"	"	"	"
J5	1S1 "	"	14	"	"	"	"	"
J6	1S0 "	"	35	"	"	"	"	"
J7	$1C_{n+4}$ "	"	19	"	"	"	"	"
J8	2S5 "	"	9	"	"	"	"	"
J9	2S4 "	"	29	"	"	"	"	"
J10	2S3 "	"	3	"	"	"	"	"
J11	2S2 "	"	4	"	"	"	"	"
J12	2S1 "	"	2	"	"	"	"	"
J13	2S0 "	"	5	"	"	"	"	"
J14	$2C_{n+4}$ "	"	8	"	"	"	"	"

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CAUTION.

Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-2 for tables VIII and X and item X01-1 for table IX.
2. With point A disconnected, item X01-1 should indicate zero current on the 1 µA range.
3. When performing the output short circuit current test, only one output shall be shorted at a time and the measurement taken within 5 seconds.

* FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX, and X.

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TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	Power supply current, low level	B	40	260	mA	11
K2	Power supply current, high level	Open	40	260	mA	11

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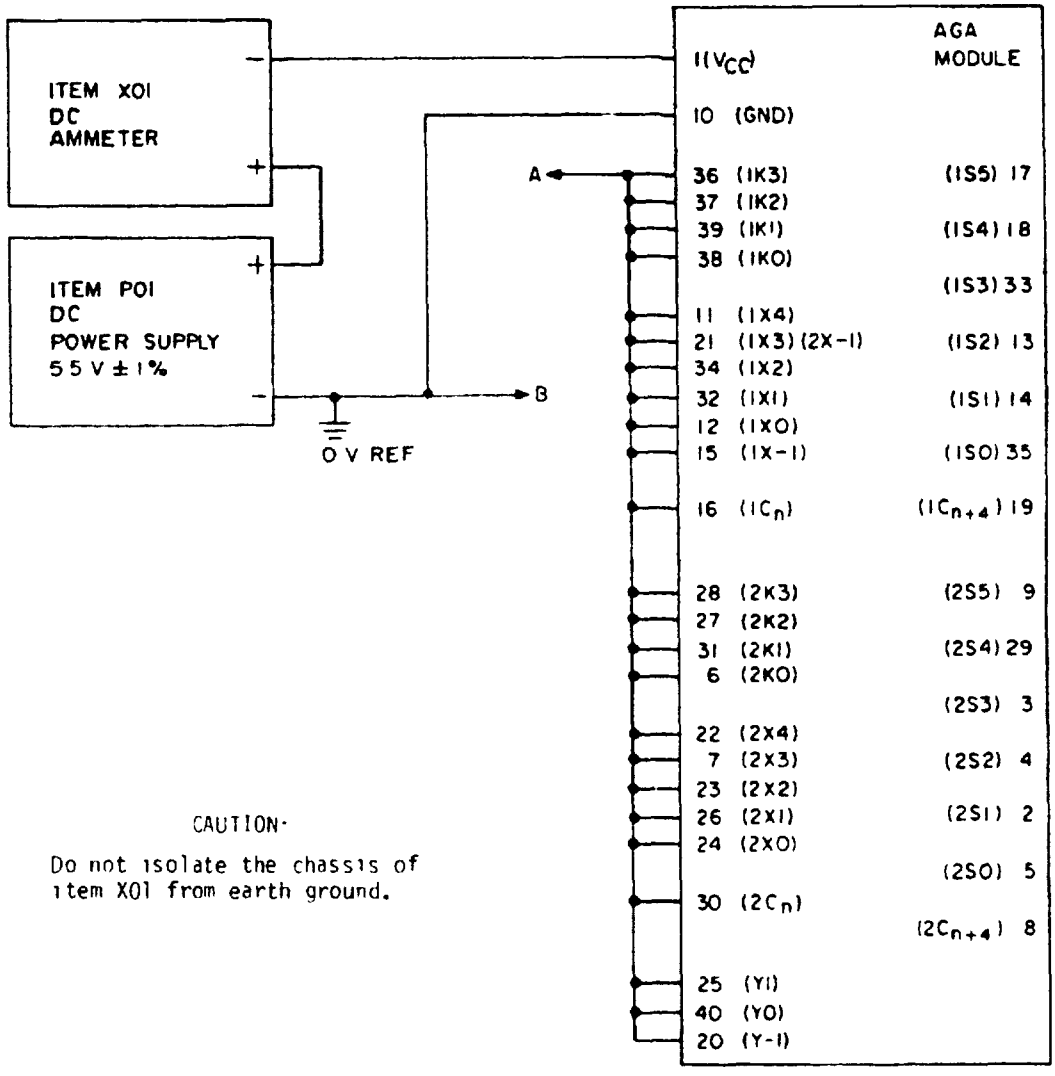


FIGURE 11 Power supply current (low-level and high-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1S0 Transition time (t_{TLH}), C_L = 75 pF	20,25,40	38	35	Open	4	30	ns	12,13
L2	1S1 "	"	39	14	"	"	"	"	"
L3	1S2 "	"	37	13	"	"	"	"	"
L4	1S3 "	"	36	33	"	"	"	"	"
L5	1S4 "	"	"	18	"	"	"	"	"
L6	1S5 "	"	"	17	"	"	"	"	"
L7	$1C_{n+4}$ "	25,40	"	19	20	"	"	"	"
L8	2S0 "	20,25,40	6	5	Open	"	"	"	"
L9	2S1 "	"	31	2	"	"	"	"	"
L10	2S2 "	"	27	4	"	"	"	"	"
L11	2S3 "	"	28	3	"	"	"	"	"
L12	2S4 "	"	"	29	"	"	"	"	"
L13	2S5 "	"	"	9	"	"	"	"	"
L14	$2C_{n+4}$ "	25,40	"	8	20	"	"	"	"

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1S0	Transition time (t_{THL}), C_L = 75 pF	20,25,40	38	35	Open	4	15	ns	12,13
M2	1S1	"	"	39	14	"	"	"	"	"
M3	1S2	"	"	37	13	"	"	"	"	"
M4	1S3	"	"	36	33	"	"	"	"	"
M5	1S4	"	"	"	18	"	"	"	"	"
M6	1S5	"	"	"	17	"	"	"	"	"
M7	$1C_{n+4}$	"	25,40	"	19	20	"	"	"	"
M8	2S0	"	20,25,40	6	5	Open	"	"	"	"
M9	2S1	"	"	31	2	"	"	"	"	"
M10	2S2	"	"	27	4	"	"	"	"	"
M11	2S3	"	"	28	3	"	"	"	"	"
M12	2S4	"	"	"	29	"	"	"	"	"
M13	2S5	"	"	"	9	"	"	"	"	"
M14	$2C_{n+4}$	"	25,40	"	8	20	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	$1C_n$ to $1C_{n+4}$ Propagation delay time (t_{PHL}), C_L = 75 pF	S	36,37,38,39,40	16	19	11,12,15,20,21, 25,32,34	7	27	ns	12,13
N2	$2C_n$ to $2C_{n+4}$ "	S	6,27,28,31,40	30	8	7,20,21,22,23, 24,25,26	"	27	"	12,13
N3	$1C_n$ to 1S0 "	S	36,37,38,39,40	16	35	11,12,15,20,21, 25,32,34	"	29	"	12,14
N4	$1C_n$ to 1S1 "	S	"	"	14	"	"	"	"	"
N5	$1C_n$ to 1S2 "	Q	"	"	13	"	"	"	"	"
N6	$1C_n$ to 1S3 "	Q	"	"	33	"	"	"	"	"
N7	$2C_n$ to 2S0 "	S	6,27,28,31,40	30	5	7,20,21,22,23, 24,25,26	"	"	"	"
N8	$2C_n$ to 2S1 "	S	"	"	2	"	"	"	"	"
N9	$2C_n$ to 2S2 "	Q	"	"	4	"	"	"	"	"
N10	$2C_n$ to 2S3 "	Q	"	"	3	"	"	"	"	"
N11	$1C_n$ to 1S4 "	S	36,37,38,39,40	16	18	11,12,15,20,21, 25,32,34	8	32	"	"
N12	$1C_n$ to 1S5 "	S	36,37,38,39,40	16	17	11,12,15,20,21, 25,32,34	"	"	"	"
N13	$2C_n$ to 2S4 "	S	6,27,28,31,40	30	29	7,20,21,22,23, 24,25,26	"	"	"	"
N14	$2C_n$ to 2S5 "	S	6,27,28,31,40	30	9	7,20,21,22,23, 24,25,26	"	"	"	"
N15	1K0 to $1C_{n+4}$ "	S	16,36,37,39,40	38	19	11,12,15,20,21, 25,32,34	7	29	"	12,13
N16	1K1 to $1C_{n+4}$ "	Q	16,36,37,38,40	39	"	"	"	"	"	"
N17	1K2 to $1C_{n+4}$ "	Q	16,36,38,39,40	37	"	"	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N18	1K3 to $1C_{n+4}$ Propagation delay time (t_{PHL}), C_L = 75 pF	S	16,37,38,39,40	36	19	11,12,15,20,21, 25,32,34	7	29	ns	12,13
N19	2K0 to $2C_{n+4}$ "	S	27,28,30,31,40	6	8	7,20,21,22,23, 24,25,26	"	"	"	"
N20	2K1 to $2C_{n+4}$ "	Q	6,27,28,30,40	31	"	"	"	"	"	"
N21	2K2 to $2C_{n+4}$ "	Q	6,28,30,31,40	27	"	"	"	"	"	"
N22	2K3 to $2C_{n+4}$ "	S	6,27,30,31,40	28	"	"	"	"	"	"
N23	1K0 to 1S0 "	S	16,36,37,39,40	38	35	11,12,15,20,21, 25,32,34	"	"	"	12,14
N24	1K0 to 1S1 "	S	"	"	14	"	"	"	"	"
N25	1K0 to 1S2 "	Q	"	"	13	"	"	"	"	"
N26	1K0 to 1S3 "	Q	"	"	33	"	"	"	"	"
N27	2K0 to 2S0 "	S	27,28,30,31,40	6	5	7,20,21,22,23, 24,25,26	"	"	"	"
N28	2K0 to 2S1 "	S	"	"	2	"	"	"	"	"
N29	2K0 to 2S2 "	Q	"	"	4	"	"	"	"	"
N30	2K0 to 2S3 "	Q	"	"	3	"	"	"	"	"
N31	1K1 to 1S1 "	Q	16,36,37,38,40	39	14	11,12,15,20,21, 25,32,34	"	"	"	"
N32	1K1 to 1S2 "	Q	"	39	13	"	"	"	"	"
N33	1K1 to 1S3 "	Q	"	39	33	"	"	"	"	"
N34	2K1 to 2S1 "	Q	6,27,28,30,40	31	2	7,20,21,22,23, 24,25,26	"	"	"	"
N35	2K1 to 2S2 "	Q	"	31	4	"	"	"	"	"
N36	2K1 to 2S3 "	Q	"	31	3	"	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C +5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
N37	1K2 to 1S2	Propagation delay time (t_{PHL}), C_L = 75 pF	Q	16,36,38,39,40	37	13	11,12,15,20,21, 25,32,34	7	29	ns	12,14
N38	1K2 to 1S3	"	Q	16,36,38,39,40	37	33	11,12,15,20,21, 25,32,34	"	"	"	"
N39	2K2 to 2S2	"	Q	6,28,30,31,40	27	4	7,20,21,22,23, 24,25,26	"	"	"	"
N40	2K2 to 2S3	"	Q	6,28,30,31,40	27	3	7,20,21,22,23, 24,25,26	"	"	"	"
N41	1K3 to 1S3	"	S	16,37,38,39,40	36	33	11,12,15,20,21, 25,32,34	"	"	"	"
N42	2K3 to 2S3	"	S	6,27,30,31,40	28	3	7,20,21,22,23, 24,25,26	"	"	"	"
N43	1K0 to 1S4	"	Q	16,36,37,39,40	38	18	11,12,15,20,21, 25,32,34	10	36	"	"
N44	1K0 to 1S5	"	Q	16,36,37,39,40	38	17	"	"	"	"	"
N45	1K1 to 1S4	"	S	16,36,37,39,40	39	18	"	"	"	"	"
N46	1K1 to 1S5	"	Q	16,36,37,39,40	39	17	"	"	"	"	"
N47	1K2 to 1S4	"	S	16,36,38,39,40	37	18	"	"	"	"	"
N48	1K2 to 1S5	"	Q	16,36,38,39,40	37	17	"	"	"	"	"
N49	2K0 to 2S4	"	Q	27,28,30,31,40	6	29	7,20,21,22,23, 24,25,26	"	"	"	"
N50	2K0 to 2S5	"	Q	27,28,30,31,40	6	9	"	"	"	"	"
N51	2K1 to 2S4	"	S	6,27,28,30,40	31	29	"	"	"	"	"
N52	2K1 to 2S5	"	Q	6,27,28,30,40	31	9	"	"	"	"	"
N53	2K2 to 2S4	"	S	6,28,30,31,40	27	29	"	"	"	"	"
N54	2K2 to 2S5	"	Q	6,28,30,31,40	27	9	"	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
N55	1K3 to 1S4	Propagation delay time (t_{PHL}), C_L = 75 pF	S	37,38,39,40	36	18	11,12,15,16,20, 21,25,32,34	10	36	ns	12,13
N56	1K3 to 1S5	"	S	37,38,39,40	36	17	11,12,15,16,20, 21,25,32,34	"	"	"	"
N57	2K3 to 2S4	"	S	6,27,31,40	28	29	7,20,21,22,23, 24,25,26,30	"	"	"	"
N58	2K3 to 2S5	"	S	6,27,31,40	28	9	7,20,21,22,23, 24,25,26,30	"	"	"	"
N59	1X-1 to $1C_{n+4}$	"	S	11,12,16,20,21, 32,34,40	15	19	25,36,37,38,39	11	37	"	"
N60	2X-1 to $2C_{n+4}$	"	S	7,20,22,23,24, 26,30,40	21	8	6,25,27,28,31	"	"	"	"
N61	1X0 to $1C_{n+4}$	"	Q	11,15,16,21,32, 34,40	12	19	20,25,36,37,38, 39	"	"	"	"
N62	1X1 to $1C_{n+4}$	"	Q	11,12,15,16,21, 34,40	32	19	"	"	"	"	"
N63	1X2 to $1C_{n+4}$	"	Q	11,12,15,16,21, 32,40	34	19	"	"	"	"	"
N64	2X0 to $2C_{n+4}$	"	Q	7,21,22,23,26, 30,40	24	8	6,20,25,27,28,31	"	"	"	"
N65	2X1 to $2C_{n+4}$	"	Q	7,21,22,23,24, 30,40	26	"	"	"	"	"	"
N66	2X2 to $2C_{n+4}$	"	Q	7,21,22,24,26, 30,40	23	"	"	"	"	"	"
N67	1X3 to $1C_{n+4}$	"	S	11,12,15,16,32, 34,40	21	19	20,25,36,37,38, 39	"	"	"	"
N68	2X3 to $2C_{n+4}$	"	S	21,22,23,24,26, 30,40	7	8	6,20,25,27,28,31	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
N69	1X-1 to 1S0	Propagation delay time (t_{PHL}), C_L = 75 pF	Q	11,12,16,20,21, 32,34,40	15	35	25,36,37,38,39	12	44	ns	12,14
N70	1X-1 to 1S1	"	Q	"	"	14	"	"	"	"	"
N71	1X-1 to 1S2	"	Q	"	"	13	"	"	"	"	"
N72	1X-1 to 1S3	"	Q	"	"	33	"	"	"	"	"
N73	2X-1 to 2S0	"	Q	7,20,22,23,24,26, 30,40	21	5	6,25,27,28,31	"	"	"	"
N74	2X-1 to 2S1	"	Q	"	"	2	"	"	"	"	"
N75	2X-1 to 2S2	"	Q	"	"	4	"	"	"	"	"
N76	2X-1 to 2S3	"	Q	"	"	3	"	"	"	"	"
N77	1X0 to 1S0	"	S	11,15,16,21,32, 34,40	12	35	20,25,36,37,38, 39	"	"	"	"
N78	1X0 to 1S1	"	Q	"	"	14	"	"	"	"	"
N79	1X0 to 1S2	"	Q	"	"	13	"	"	"	"	"
N80	1X0 to 1S3	"	Q	"	"	33	"	"	"	"	"
N81	2X0 to 2S0	"	S	7,21,22,23,26, 30,40	24	5	6,20,25,27,28,31	"	"	"	"
N82	2X0 to 2S1	"	Q	"	"	2	"	"	"	"	"
N83	2X0 to 2S2	"	Q	"	"	4	"	"	"	"	"
N84	2X0 to 2S3	"	Q	"	"	3	"	"	"	"	"
N85	1X1 to 1S1	"	S	11,12,15,16,21, 34,40	32	14	20,25,36,37,38, 39	"	"	"	"
N86	1X1 to 1S2	"	Q	"	"	13	"	"	"	"	"
N87	1X1 to 1S3	"	Q	"	"	33	"	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
N88	2X1 to 2S1	Propagation delay time (t_{PHL}), C_L = 75 pF	S	7,21,22,23,24, 30,40	26	2	6,20,25,27,28,31	12	44	ns	12,14
N89	2X1 to 2S2	"	Q	"	"	4	"	"	"	"	"
N90	2X1 to 2S3	"	Q	"	"	3	"	"	"	"	"
N91	1X2 to 1S2	"	Q	11,12,15,16,21, 32,40	34	13	20,25,36,37,38, 39	"	"	"	"
N92	1X2 to 1S3	"	Q	11,12,15,16,21, 32,40	34	33	20,25,36,37,38, 39	"	"	"	"
N93	2X2 to 2S2	"	Q	7,21,22,24,26, 30,40	23	4	6,20,25,27,28,31	"	"	"	"
N94	2X2 to 2S3	"	Q	7,21,22,24,26, 30,40	23	3	6,20,25,27,28,31	"	"	"	"
N95	1X3 to 1S3	"	S	11,12,15,16,32, 34,40	21	33	20,25,36,37,38, 39	"	"	"	"
N96	2X3 to 2S3	"	S	21,22,23,24,26, 30,40	7	3	6,20,25,27,28,31	"	"	"	"
N97	1X-1 to 1S4	"	Q	11,12,16,20,21, 32,34,40	15	18	25,36,37,38,39	13	"	"	"
N98	1X-1 to 1S5	"	Q	11,12,16,20,21, 32,34,40	15	17	25,36,37,38,39	"	"	"	"
N99	2X-1 to 2S4	"	Q	7,20,22,23,24, 26,30,40	21	29	6,25,27,28,31	"	"	"	"
N100	2X-1 to 2S5	"	Q	7,20,22,23,24, 26,30,40	21	9	6,25,27,28,31	"	"	"	"
N101	1X0 to 1S4	"	Q	11,15,16,21,32, 34,40	12	18	20,25,36,37,38, 39	"	"	"	"
N102	1X0 to 1S5	"	Q	11,15,16,21,32, 34,40	12	17	20,25,36,37,38, 39	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N103	1X1 to 1S4	Propagation delay time (t_{PHL}), C_L = 75 pF	Q	11,12,15,16,21, 34,40	32	18	20,25,36,37,38, 39	13	44	ns	12,14
N104	1X1 to 1S5	"	Q	11,12,15,16,21, 34,40	32	17	"	"	"	"	"
N105	1X2 to 1S4	"	S	11,12,15,16,21, 32,40	34	18	"	"	"	"	"
N106	1X2 to 1S5	"	S	11,12,15,16,21, 32,40	34	17	"	"	"	"	"
N107	2X0 to 2S4	"	Q	7,21,22,23,26, 30,40	24	29	6,20,25,27,28,31	"	"	"	"
N108	2X0 to 2S5	"	Q	7,21,22,23,26, 30,40	24	9	"	"	"	"	"
N109	2X1 to 2S4	"	Q	7,21,22,23,24, 30,40	26	29	"	"	"	"	"
N110	2X1 to 2S5	"	Q	7,21,22,23,24, 30,40	26	9	"	"	"	"	"
N111	2X2 to 2S4	"	S	7,21,22,24,26, 30,40	23	29	"	"	"	"	"
N112	2X2 to 2S5	"	S	7,21,22,24,26, 30,40	23	9	"	"	"	"	"
N113	1X3 to 1S4	"	S	11,12,15,32,34, 40	21	18	16,20,25,36,37, 38,39	"	"	"	12,13
N114	1X3 to 1S5	"	Q	11,12,15,32,34, 40	21	17	16,20,25,36,37, 38,39	"	"	"	"
N115	2X3 to 2S4	"	S	21,22,23,24,26, 40	7	29	6,20,25,27,28, 30,31	"	"	"	"
N116	2X3 to 2S5	"	Q	21,22,23,24,26, 40	7	9	6,20,25,27,28, 30,31	"	"	"	"

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TABLE XIV Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
N117	1X4 to 1S4	Propagation delay time (t_{PHL}), C_L = 75 pF	Q	12,15,20,21,32, 34,40	11	18	16,25,36,37,38, 39	13	44	ns	12,13
N118	1X4 to 1S5	"	S	12,15,20,21,32, 34,40	11	17	16,25,36,37,38, 39	"	"	"	"
N119	2X4 to 2S4	"	Q	7,20,21,23,24, 26,40	22	29	6,25,27,28,30,31	"	"	"	"
N120	2X4 to 2S5	"	S	7,20,21,23,24, 26,40	22	9	6,25,27,28,30,31	"	"	"	"
N121	Y-1 to $1C_{n+4}$	"	S	15,16,25	20	19	11,12,21,32,34, 36,37,38,39,40	19	59	"	12,13
N122	Y-1 to $2C_{n+4}$	"	S	21,25,30	20	8	6,7,22,23,24,26, 27,28,31,40	"	"	"	"
N123	Y0 to $1C_{n+4}$	"	S	12,15,16,20,25	40	19	11,21,32,34,36, 37,38,39	"	"	"	"
N124	Y0 to $2C_{n+4}$	"	S	20,21,24,25,30	40	8	6,7,22,23,26,27, 28,31	"	"	"	"
N125	Y1 to $1C_{n+4}$	"	S	11,12,15,16,20, 21,32,34	25	19	36,37,38,39,40	"	"	"	12,14
N126	Y1 to $2C_{n+4}$	"	S	7,20,21,22,23, 24,26,30	25	8	6,27,28,31,40	"	"	"	"
N127	Y-1 to 1S0	"	S	15,16,25	20	35	11,12,21,32,34, 36,37,38,39,40	14	49	"	"
N128	Y-1 to 1S1	"	S	"	"	14	"	"	"	"	"
N129	Y-1 to 1S2	"	Q	"	"	13	"	"	"	"	"
N130	Y-1 to 1S3	"	Q	"	"	33	"	"	"	"	"
N131	Y-1 to 2S0	"	S	21,25,30	"	5	6,7,22,23,24,26, 27,28,31,40	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
N132	Y-1 to 2S1	Propagation delay time (t_{PHL}), C_L = 75 pF	S	21,25,30	20	2	6,7,22,23,24,26, 27,28,31,40	14	49	ns	12,14
N133	Y-1 to 2S2	"	Q	"	"	4	"	"	"	"	"
N134	Y-1 to 2S3	"	Q	"	"	3	"	"	"	"	"
N135	Y0 to 1S0	"	S	12,15,16,20,25	40	35	11,21,32,34,36, 37,38,39	"	"	"	"
N136	Y0 to 1S1	"	S	"	"	14	"	"	"	"	"
N137	Y0 to 1S2	"	Q	"	"	13	"	"	"	"	"
N138	Y0 to 1S3	"	Q	"	"	33	"	"	"	"	"
N139	Y0 to 2S0	"	S	20,21,24,25,30	"	5	6,7,22,23,26,27, 28,31	"	"	"	"
N140	Y0 to 2S1	"	S	"	"	2	"	"	"	"	"
N141	Y0 to 2S2	"	Q	"	"	4	"	"	"	"	"
N142	Y0 to 2S3	"	Q	"	"	3	"	"	"	"	"
N143	Y1 to 1S0	"	S	11,12,15,20,21, 32,34	25	35	16,36,37,38,39, 40	"	"	"	"
N144	Y1 to 1S1	"	Q	"	"	14	"	"	"	"	"
N145	Y1 to 1S2	"	Q	"	"	13	"	"	"	"	"
N146	Y1 to 1S3	"	S	"	"	33	"	"	"	"	"
N147	Y1 to 2S0	"	S	7,20,21,22,23, 24,26	"	5	6,27,28,30,31,40	"	"	"	"
N148	Y1 to 2S1	"	Q	"	"	2	"	"	"	"	"
N149	Y1 to 2S2	"	Q	"	"	4	"	"	"	"	"
N150	Y1 to 2S3	"	S	"	"	3	"	"	"	"	"

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
N151	Y-1 to 1S4	Propagation delay time (t_{PHL}), C_L = 75 pF	S	15,16,25	20	18	11,12,21,32,34, 36,37,38,39,40	16	51	ns	12,14
N152	Y-1 to 1S5	"	S	15,16,25	"	17	11,12,21,32,34, 36,37,38,39,40	"	"	"	"
N153	Y-1 to 2S4	"	S	21,25,30	"	29	6,7,22,23,24,26, 27,28,31,40	"	"	"	"
N154	Y-1 to 2S5	"	S	21,25,30	"	9	6,7,22,23,24,26, 27,28,31,40	"	"	"	"
N155	Y0 to 1S4	"	S	12,15,16,20,25	40	18	11,21,32,34,36, 37,38,39	"	"	"	"
N156	Y0 to 1S5	"	S	12,15,16,20,25	"	17	11,21,32,34,36, 37,38,39	"	"	"	"
N157	Y0 to 2S4	"	S	20,21,24,25,30	"	29	6,7,22,23,26,27, 28,31	"	"	"	"
N158	Y0 to 2S5	"	S	20,21,24,25,30	"	9	6,7,22,23,26,27, 28,31	"	"	"	"
N159	Y1 to 1S4	"	S	11,12,15,20,21, 32,34,	25	18	16,36,37,38,39, 40	"	"	"	"
N160	Y1 to 1S5	"	S	11,12,15,20,21, 32,34,	"	17	16,36,37,38,39, 40	"	"	"	"
N161	Y1 to 2S4	"	S	7,20,21,22,23, 24,26,	"	29	6,27,28,30,31,40	"	"	"	"
N162	Y1 to 2S5	"	S	7,20,21,22,23, 24,26,	"	9	6,27,28,30,31,40	"	"	"	"

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TABLE XV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	$1C_n$ to $1C_{n+4}$	Propagation delay time (t_{PLH}), C_L = 75 pF	S	36,37,38,39,40	16	19	11,12,15,20,21, 25,32,34	6	26	ns	12,13
P2	$2C_n$ to $2C_{n+4}$	"	S	6,27,28,31,40	30	8	7,20,21,22,23, 24,25,26	6	26	"	12,13
P3	$1C_n$ to 1S0	"	S	36,37,38,39,40	16	35	11,12,15,20,21, 25,32,34	9	33	"	12,14
P4	$1C_n$ to 1S1	"	S	"	"	14	"	"	"	"	"
P5	$1C_n$ to 1S2	"	Q	"	"	13	"	"	"	"	"
P6	$1C_n$ to 1S3	"	Q	"	"	33	"	"	"	"	"
P7	$2C_n$ to 2S0	"	S	6,27,28,31,40	30	5	7,20,21,22,23, 24,25,26	"	"	"	"
P8	$2C_n$ to 2S1	"	S	"	"	2	"	"	"	"	"
P9	$2C_n$ to 2S2	"	Q	"	"	4	"	"	"	"	"
P10	$2C_n$ to 2S3	"	Q	"	"	3	"	"	"	"	"
P11	$1C_n$ to 1S4	"	S	36,37,38,39,40	16	18	11,12,15,20,21, 25,32,34	11	39	"	"
P12	$1C_n$ to 1S5	"	S	36,37,38,39,40	16	17	11,12,15,20,21, 25,32,34	"	"	"	"
P13	$2C_n$ to 2S4	"	S	6,27,28,31,40	30	29	7,20,21,22,23, 24,25,26	"	"	"	"
P14	$2C_n$ to 2S5	"	S	6,27,28,31,40	30	9	7,20,21,22,23, 24,25,26	"	"	"	"
P15	1K0 to $1C_{n+4}$	"	S	16,36,37,39,40	38	19	11,12,15,20,21, 25,32,34	6	24	"	12,13
P16	1K1 to $1C_{n+4}$	"	Q	16,36,37,38,40	39	"	"	"	"	"	"
P17	1K2 to $1C_{n+4}$	"	Q	16,36,38,39,40	37	"	"	"	"	"	"

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TABLE XV Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P18	1K3 to $1C_{n+4}$ Propagation delay time (t_{PLH}), C_L = 75 pF	S	16,37,38,39,40	36	19	11,12,15,20,21, 25,32,34	6	24	ns	12,13
P19	2K0 to $2C_{n+4}$ "	S	27,28,30,31,40	6	8	7,20,21,22,23, 24,25,26	"	"	"	"
P20	2K1 to $2C_{n+4}$ "	Q	6,27,28,30,40	31	"	"	"	"	"	"
P21	2K2 to $2C_{n+4}$ "	Q	6,28,30,31,40	27	"	"	"	"	"	"
P22	2K3 to $2C_{n+4}$ "	S	6,27,30,31,40	28	"	"	"	"	"	"
P23	1K0 to 1S0 "	S	16,36,37,39,40	38	35	11,12,15,20,21, 25,32,34	9	33	"	12,14
P24	1K0 to 1S1 "	S	"	"	14	"	"	"	"	"
P25	1K0 to 1S2 "	Q	"	"	13	"	"	"	"	"
P26	1K0 to 1S3 "	Q	"	"	33	"	"	"	"	"
P27	2K0 to 2S0 "	S	27,28,30,31,40	6	5	7,20,21,22,23, 24,25,26	"	"	"	"
P28	2K0 to 2S1 "	S	"	"	2	"	"	"	"	"
P29	2K0 to 2S2 "	Q	"	"	4	"	"	"	"	"
P30	2K0 to 2S3 "	Q	"	"	3	"	"	"	"	"
P31	1K1 to 1S1 "	Q	16,36,37,38,40	39	14	11,12,15,20,21, 25,32,34	"	"	"	"
P32	1K1 to 1S2 "	Q	"	39	13	"	"	"	"	"
P33	1K1 to 1S3 "	Q	"	39	33	"	"	"	"	"
P34	2K1 to 2S1 "	Q	6,27,28,30,40	31	2	7,20,21,22,23, 24,25,26	"	"	"	"
P35	2K1 to 2S2 "	Q	"	31	4	"	"	"	"	"
P36	2K1 to 2S3 "	Q	"	31	3	"	"	"	"	"

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TABLE XV Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
P37	1K2 to 1S2	Propagation delay time (t_{PLH}), C_L = 75 pF	Q	16,36,38,39,40	37	13	11,12,15,20,21, 25,32,34	9	33	ns	12,14
P38	1K2 to 1S3	"	Q	16,36,38,39,40	37	33	11,12,15,20,21, 25,32,34	"	"	"	"
P39	2K2 to 2S2	"	Q	6,28,30,31,40	27	4	7,20,21,22,23, 24,25,26	"	"	"	"
P40	2K2 to 2S3	"	Q	6,28,30,31,40	27	3	7,20,21,22,23, 24,25,26	"	"	"	"
P41	1K3 to 1S3	"	S	16,37,38,39,40	36	33	11,12,15,20,21, 25,32,34	"	"	"	"
P42	2K3 to 2S3	"	S	6,27,30,31,40	28	3	7,20,21,22,23, 24,25,26	"	"	"	"
P43	1K0 to 1S4	"	Q	16,36,37,39,40	38	18	11,12,15,20,21, 25,32,34	"	39	"	"
P44	1K0 to 1S5	"	Q	16,36,37,39,40	38	17	"	"	"	"	"
P45	1K1 to 1S4	"	S	16,36,37,39,40	39	18	"	"	"	"	"
P46	1K1 to 1S5	"	Q	16,36,37,39,40	39	17	"	"	"	"	"
P47	1K2 to 1S4	"	S	16,36,38,39,40	37	18	"	"	"	"	"
P48	1K2 to 1S5	"	Q	16,36,38,39,40	37	17	"	"	"	"	"
P49	2K0 to 2S4	"	Q	27,28,30,31,40	6	29	7,20,21,22,23, 24,25,26	"	"	"	"
P50	2K0 to 2S5	"	Q	27,28,30,31,40	6	9	"	"	"	"	"
P51	2K1 to 2S4	"	S	6,27,28,30,40	31	29	"	"	"	"	"
P52	2K1 to 2S5	"	Q	6,27,28,30,40	31	9	"	"	"	"	"
P53	2K2 to 2S4	"	S	6,28,30,31,40	27	29	"	"	"	"	"
P54	2K2 to 2S5	"	Q	6,28,30,31,40	27	9	"	"	"	"	"

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P55	1K3 to 1S4 Propagation delay time (t_{PLH}), C_L = 75 pF	S	37,38,39,40	36	18	11,12,15,16,20, 21,25,32,34	9	39	ns	12,13
P56	1K3 to 1S5 "	S	37,38,39,40	36	17	11,12,15,16,20, 21,25,32,34	"	"	"	"
P57	1K3 to 2S4 "	S	6,27,31,40	28	29	7,20,21,22,23, 24,25,26,30	"	"	"	"
P58	2K3 to 2S5 "	S	6,27,31,40	28	9	7,20,21,22,23, 24,25,26,30	"	"	"	"
P59	1X-1 to $1C_{n+4}$ "	S	11,12,16,20,21, 32,34,40	15	19	25,36,37,38,39	10	38	"	"
P60	2X-1 to $2C_{n+4}$ "	S	7,20,22,23,24, 26,30,40	21	8	6,25,27,28,31	"	"	"	"
P61	1X0 to $1C_{n+4}$ "	Q	11,15,16,21,32, 34,40	12	19	20,25,36,37,38, 39	"	"	"	"
P62	1X1 to $1C_{n+4}$ "	Q	11,12,15,16,21, 34,40	32	19	"	"	"	"	"
P63	1X2 to $1C_{n+4}$ "	Q	11,12,15,16,21, 32,40	34	19	"	"	"	"	"
P64	2X0 to $2C_{n+4}$ "	Q	7,21,22,23,26, 30,40	24	8	6,20,25,27,28,31	"	"	"	"
P65	2X1 to $2C_{n+4}$ "	Q	7,21,22,23,24, 30,40	26	"	"	"	"	"	"
P66	2X2 to $2C_{n+4}$ "	Q	7,21,22,24,26, 30,40	23	"	"	"	"	"	"
P67	1X3 to $1C_{n+4}$ "	S	11,12,15,16,32, 34,40	21	19	20,25,36,37,38, 39	"	"	"	"
P68	2X3 to $2C_{n+4}$ "	S	21,22,23,24,26, 30,40	7	8	6,20,25,27,28,31	"	"	"	"

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
P69	1X-1 to 1S0	Propagation delay time (t_{PLH}), C_L = 75 pF	Q	11,12,16,20,21, 32,34,40	15	35	25,36,37,38,39	13	43	ns	12,14
P70	1X-1 to 1S1	"	Q	"	"	14	"	"	"	"	"
P71	1X-1 to 1S2	"	Q	"	"	13	"	"	"	"	"
P72	1X-1 to 1S3	"	Q	"	"	33	"	"	"	"	"
P73	2X-1 to 2S0	"	Q	7,20,22,23,24,26, 30,40	21	5	6,25,27,28,31	"	"	"	"
P74	2X-1 to 2S1	"	Q	"	"	2	"	"	"	"	"
P75	2X-1 to 2S2	"	Q	"	"	4	"	"	"	"	"
P76	2X-1 to 2S3	"	Q	"	"	3	"	"	"	"	"
P77	1X0 to 1S0	"	S	11,15,16,21,32, 34,40	12	35	20,25,36,37,38, 39	"	"	"	"
P78	1X0 to 1S1	"	Q	"	"	14	"	"	"	"	"
P79	1X0 to 1S2	"	Q	"	"	13	"	"	"	"	"
P80	1X0 to 1S3	"	Q	"	"	33	"	"	"	"	"
P81	2X0 to 2S0	"	S	7,21,22,23,26, 30,40	24	5	6,20,25,27,28,31	"	"	"	"
P82	2X0 to 2S1	"	Q	"	"	2	"	"	"	"	"
P83	2X0 to 2S2	"	Q	"	"	4	"	"	"	"	"
P84	2X0 to 2S3	"	Q	"	"	3	"	"	"	"	"
P85	1X1 to 1S1	"	S	11,12,15,16,21, 34,40	32	14	20,25,36,37,38, 39	"	"	"	"
P86	1X1 to 1S2	"	Q	"	"	13	"	"	"	"	"
P87	1X1 to 1S3	"	Q	"	"	33	"	"	"	"	"

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
P88	2X1 to 2S1	Propagation delay time (t_{PLH}), C_L = 75 pF	S	7,21,22,23,24, 30,40	26	2	6,20,25,27,28,31	13	43	ns	12,14
P89	2X1 to 2S2	"	Q	"	"	4	"	"	"	"	"
P90	2X1 to 2S3	"	Q	"	"	3	"	"	"	"	"
P91	1X2 to 1S2	"	Q	11,12,15,16,21, 32,40	34	13	20,25,36,37,38, 39	"	"	"	"
P92	1X2 to 1S3	"	Q	11,12,15,16,21, 32,40	34	33	20,25,36,37,38, 39	"	"	"	"
P93	2X2 to 2S2	"	Q	7,21,22,24,26, 30,40	23	4	6,20,25,27,28,31	"	"	"	"
P94	2X2 to 2S3	"	Q	7,21,22,24,26, 30,40	23	3	6,20,25,27,28,31	"	"	"	"
P95	1X3 to 1S3	"	S	11,12,15,16,32, 34,40	21	33	20,25,36,37,38, 39	"	"	"	"
P96	2X3 to 2S3	"	S	21,22,23,24,26, 30,40	7	3	6,20,25,27,28,31	"	"	"	"
P97	1X-1 to 1S4	"	Q	11,12,16,20,21, 32,34,40	15	18	25,36,37,38,39	16	51	"	"
P98	1X-1 to 1S5	"	Q	11,12,16,20,21, 32,34,40	15	17	25,36,37,38,39	"	"	"	"
P99	2X-1 to 2S4	"	Q	7,20,22,23,24, 26,30,40	21	29	6,25,27,28,31	"	"	"	"
P100	2X-1 to 2S5	"	Q	7,20,22,23,24, 26,30,40	21	9	6,25,27,28,31	"	"	"	"
P101	1X0 to 1S4	"	Q	11,15,16,21,32, 34,40	12	18	20,25,36,37,38, 39	"	"	"	"
P102	1X0 to 1S5	"	Q	11,15,16,21,32, 34,40	12	17	20,25,36,37,38, 39	"	"	"	"

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TABLE XV Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P103	1X1 to 1S4	Propagation delay time (t_{PLH}), C_L = 75 pF	Q	11,12,15,16,21, 34,40	32	18	20,25,36,37,38, 39	16	51	ns	12,14
P104	1X1 to 1S5	"	Q	11,12,15,16,21, 34,40	32	17	"	"	"	"	"
P105	1X2 to 1S4	"	S	11,12,15,16,21, 32,40	34	18	"	"	"	"	"
P106	1X2 to 1S5	"	S	11,12,15,16,21, 32,40	34	17	"	"	"	"	"
P107	2X0 to 2S4	"	Q	7,21,22,23,26, 30,40	24	29	6,20,25,27,28,31	"	"	"	"
P108	2X0 to 2S5	"	Q	7,21,22,23,26, 30,40	24	9	"	"	"	"	"
P109	2X1 to 2S4	"	Q	7,21,22,23,24, 30,40	26	29	"	"	"	"	"
P110	2X1 to 2S5	"	Q	7,21,22,23,24, 30,40	26	9	"	"	"	"	"
P111	2X2 to 2S4	"	S	7,21,22,24,26, 30,40	23	29	"	"	"	"	"
P112	2X2 to 2S5	"	S	7,21,22,24,26, 30,40	23	9	"	"	"	"	"
P113	1X3 to 1S4	"	S	11,12,15,32,34, 40	21	18	16,20,25,36,37, 38,39	"	"	"	12,13
P114	1X3 to 1S5	"	Q	11,12,15,32,34, 40	21	17	16,20,25,36,37, 38,39	"	"	"	"
P115	2X3 to 2S4	"	S	21,22,23,24,26, 40	7	29	6,20,25,27,28, 30,31	"	"	"	"
P116	2X3 to 2S5	"	Q	21,22,23,24,26, 40	7	9	6,20,25,27,28, 30,31	"	"	"	"

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TABLE XV Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
P117	1X4 to 1S4	Propagation delay time (t_{PLH}), C_L = 75 pF	Q	12,15,20,21,32, 34,40	11	18	16,25,36,37,38, 39	16	51	ns	12,13
P118	1X4 to 1S5	"	S	12,15,20,21,32, 34,40	11	17	16,25,36,37,38, 39	"	"	"	"
P119	2X4 to 2S4	"	Q	7,20,21,23,24, 26,40	22	29	6,25,27,28,30,31	"	"	"	"
P120	2X4 to 2S5	"	S	7,20,21,23,24, 26,40	22	9	6,25,27,28,30,31	"	"	"	"
P121	Y-1 to $1C_{n+4}$	"	S	15,16,25	20	19	11,12,21,32,34, 36,37,38,39,40	14	49	"	"
P122	Y-1 to $2C_{n+4}$	"	S	21,25,30	20	8	6,7,22,23,24,26, 27,28,31,40	"	"	"	"
P123	Y0 to $1C_{n+4}$	"	S	12,15,16,20,25	40	19	11,21,32,34,36, 37,38,39	"	"	"	"
P124	Y0 to $2C_{n+4}$	"	S	20,21,24,25,30	40	8	6,7,22,23,26,27, 28,31	"	"	"	"
P125	Y1 to $1C_{n+4}$	"	S	11,12,15,16,20, 21,32,34	25	19	36,37,38,39,40	"	"	"	12,14
P126	Y1 to $2C_{n+4}$	"	S	7,20,21,22,23, 24,26,30	25	8	6,27,28,31,40	"	"	"	"
P127	Y-1 to 1S0	"	S	15,16,25	20	35	11,12,21,32,34, 36,37,38,39,40	20	64	"	"
P128	Y-1 to 1S1	"	S	"	"	14	"	"	"	"	"
P129	Y-1 to 1S2	"	Q	"	"	13	"	"	"	"	"
P130	Y-1 to 1S3	"	Q	"	"	33	"	"	"	"	"
P131	Y-1 to 2S0	"	S	21,25,30	"	5	6,7,22,23,24,26, 27,28,31,40	"	"	"	"

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

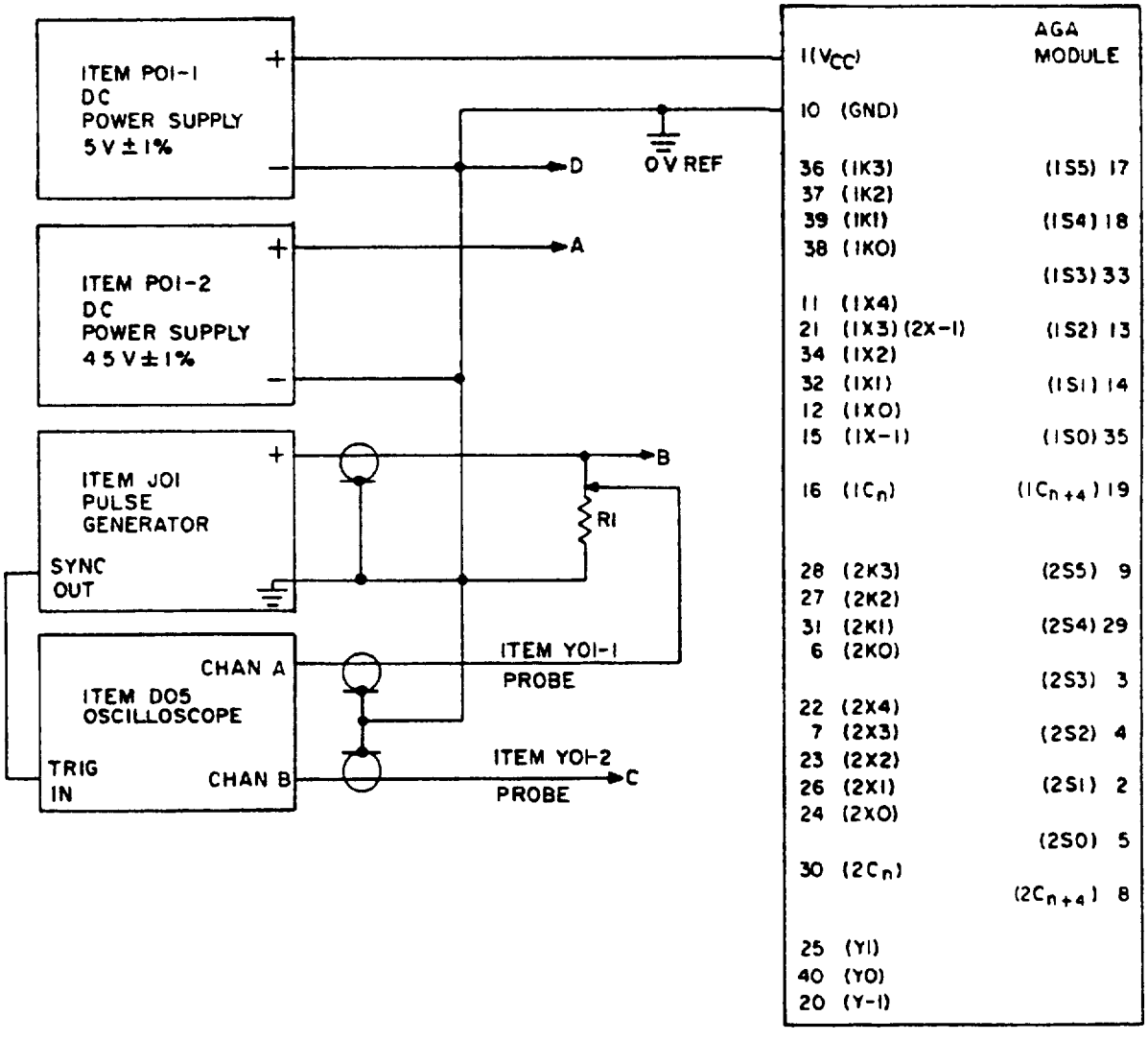
Test condition	Parameter		Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	Min	Max		
P132	Y-1 to 2S1	Propagation delay time (t_{PLH}), C_L = 75 pF	S	21,25,30	20	2	6,7,22,23,24,26, 27,28,31,40	20	64	ns	12,14
P133	Y-1 to 2S2	"	Q	"	"	4	"	"	"	"	"
P134	Y-1 to 2S3	"	Q	"	"	3	"	"	"	"	"
P135	Y0 to 1S0	"	S	12,15,16,20,25	40	35	11,21,32,34,36, 37,38,39	"	"	"	"
P136	Y0 to 1S1	"	S	"	"	14	"	"	"	"	"
P137	Y0 to 1S2	"	Q	"	"	13	"	"	"	"	"
P138	Y0 to 1S3	"	Q	"	"	33	"	"	"	"	"
P139	Y0 to 2S0	"	S	20,21,24,25,30	"	5	6,7,22,23,26,27, 28,31	"	"	"	"
P140	Y0 to 2S1	"	S	"	"	2	"	"	"	"	"
P141	Y0 to 2S2	"	Q	"	"	4	"	"	"	"	"
P142	Y0 to 2S3	"	Q	"	"	3	"	"	"	"	"
P143	Y1 to 1S0	"	S	11,12,15,20,21, 32,34	25	35	16,36,37,38,39, 40	"	"	"	"
P144	Y1 to 1S1	"	Q	"	"	14	"	"	"	"	"
P145	Y1 to 1S2	"	Q	"	"	13	"	"	"	"	"
P146	Y1 to 1S3	"	S	"	"	33	"	"	"	"	"
P147	Y1 to 2S0	"	S	7,20,21,22,23, 24,26	"	5	6,27,28,30,31,40	"	"	"	"
P148	Y1 to 2S1	"	Q	"	"	2	"	"	"	"	"
P149	Y1 to 2S2	"	Q	"	"	4	"	"	"	"	"
P150	Y1 to 2S3	"	S	"	"	3	"	"	"	"	"

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Test type (see table I)	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P151	Y-1 to 1S4	Propagation delay time (t_{PLH}), C_L = 75 pF	S	15,16,25	20	18	11,12,21,32,34, 36,37,38,39,40	22	69	ns	12,14
P152	Y-1 to 1S5	"	S	15,16,25	"	17	11,12,21,32,34, 36,37,38,39,40	"	"	"	"
P153	Y-1 to 2S4	"	S	21,25,30	"	29	6,7,22,23,24,26, 27,28,31,40	"	"	"	"
P154	Y-1 to 2S5	"	S	21,25,30	"	9	6,7,22,23,24,26, 27,28,31,40	"	"	"	"
P155	Y0 to 1S4	"	S	12,15,16,20,25	40	18	11,21,32,34,36, 37,38,39	"	"	"	"
P156	Y0 to 1S5	"	S	12,15,16,20,25	"	17	11,21,32,34,36, 37,38,39	"	"	"	"
P157	Y0 to 2S4	"	S	20,21,24,25,30	"	29	6,7,22,23,26,27, 28,31	"	"	"	"
P158	Y0 to 2S5	"	S	20,21,24,25,30	"	9	6,7,22,23,26,27, 28,31	"	"	"	"
P159	Y1 to 1S4	"	S	11,12,15,20,21, 32,34,	25	18	16,36,37,38,39, 40	"	"	"	"
P160	Y1 to 1S5	"	S	11,12,15,20,21, 32,34,	"	17	16,36,37,38,39, 40	"	"	"	"
P161	Y1 to 2S4	"	S	7,20,21,22,23, 24,26,	"	29	6,27,28,30,31,40	"	"	"	"
P162	Y1 to 2S5	"	S	7,20,21,22,23, 24,26,	"	9	6,27,28,30,31,40	"	"	"	"

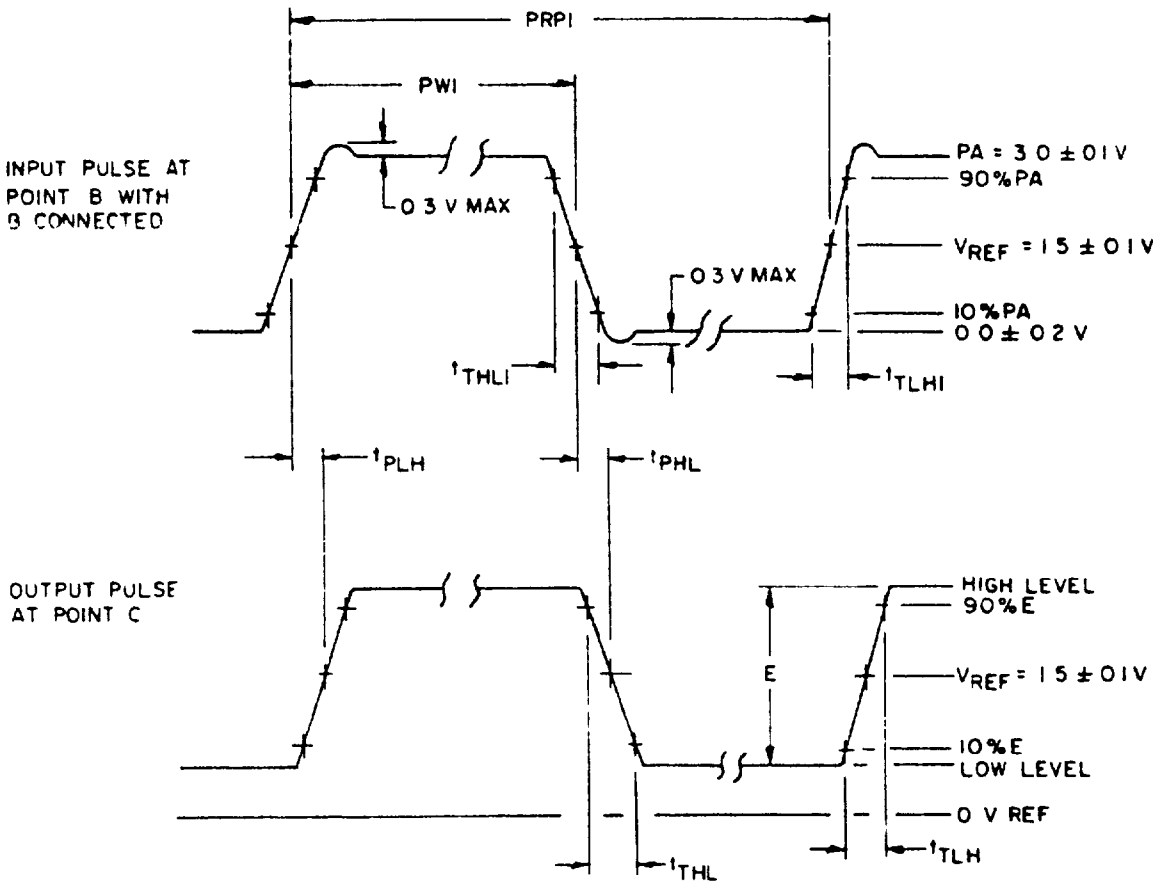
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load the output with a load as shown on figure 9.
3. Before making the measurements, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V.

* FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XII, XIII, XIV, and XV.

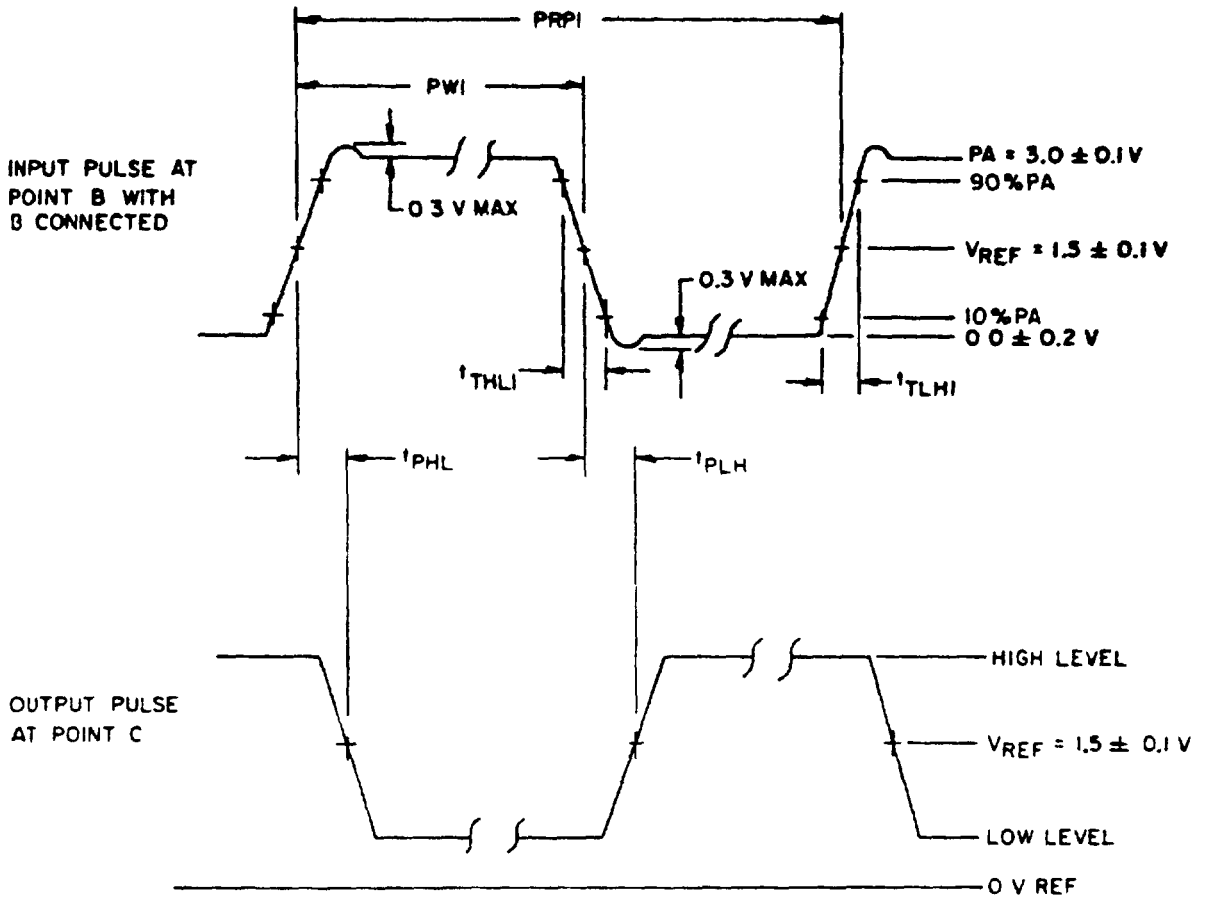
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NOTE Input pulse parameters
PWI = 300 ±30 ns
PRPI = 600 ±60 ns
$t_{TLHI} = t_{THLI} = 10 \pm 2$ ns

FIGURE 13 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XII, XIII, XIV, and XV

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NOTE Input pulse parameters
PW1 = 300 ±30 ns.
PRP1 = 600 ±60 ns
$t_{TLH1} = t_{THL1}$ = 10 ±2 ns.

FIGURE 14 Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIV and XV.

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TABLE XVI. Input clamp diode voltage test.

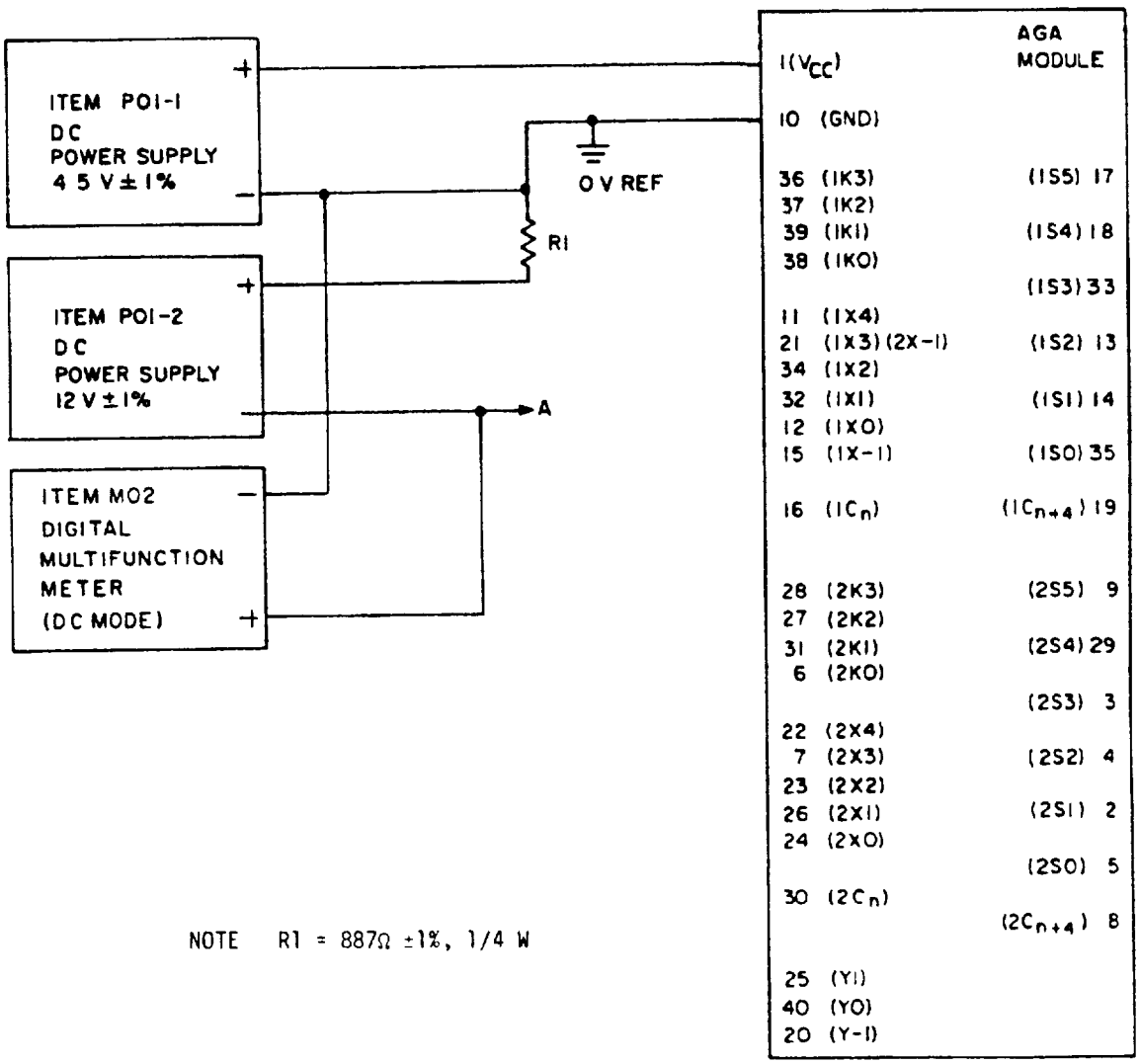
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
Q1	1K3	Input clamp diode voltage	36	-0.4	-1.5	V	15
Q2	1K2	"	37	"	"	"	"
Q3	1K1	"	39	"	"	"	"
Q4	1K0	"	38	"	"	"	"
Q5	1X4	"	11	"	"	"	"
Q6	1X2	"	34	"	"	"	"
Q7	1X1	"	32	"	"	"	"
Q8	1X0	"	12	"	"	"	"
Q9	1X-1	"	15	"	"	"	"
Q10	$1C_n$	"	16	"	"	"	"
Q11	2K3	"	28	"	"	"	"
Q12	2K2	"	27	"	"	"	"
Q13	2K1	"	31	"	"	"	"
Q14	2K0	"	6	"	"	"	"
Q15	2X4	"	22	"	"	"	"
Q16	2X3	"	7	"	"	"	"
Q17	2X2	"	23	"	"	"	"
Q18	2X1	"	26	"	"	"	"
Q19	2X0	"	24	"	"	"	"
Q20	$2C_n$	"	30	"	"	"	"
Q21	1X3,2X-1	"	21	"	"	"	"

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TABLE XVI. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
Q22	Y1	Input clamp diode voltage	25	-0.4	-1.5	V	15
Q23	Y0	"	40	"	"	"	"
Q24	Y-1	"	20	"	"	"	"

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NOTE R1 = 887Ω ±1%, 1/4 W

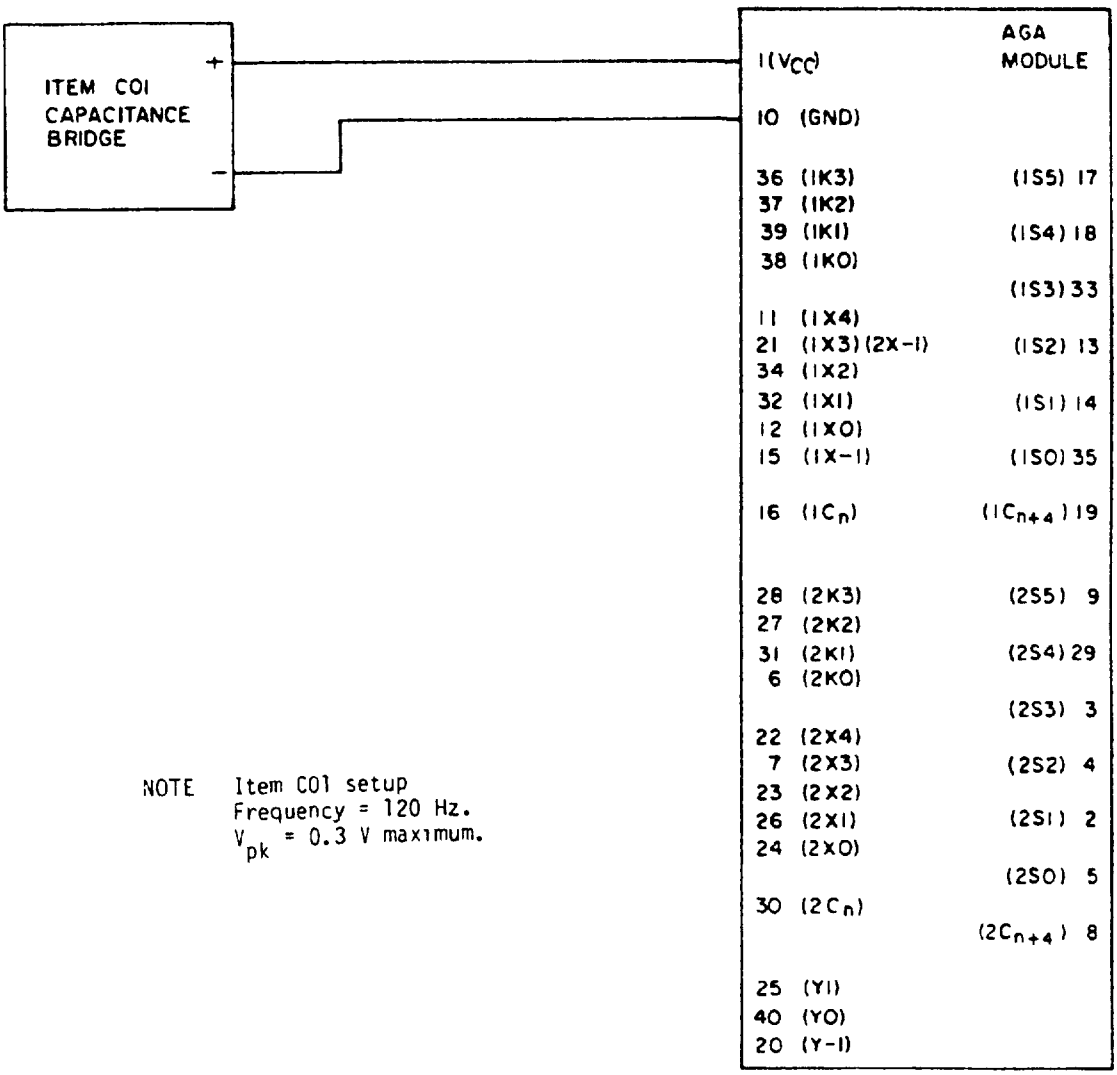
* FIGURE 15. Input clamp diode voltage test circuit for table XVI.

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TABLE XVII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply filter capacitance	7.5	11.3	µF	16

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NOTE Item C01 setup
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 16 Power supply filter capacitance test circuit for table XVII

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TABLE XVIII. Input capacitance test.

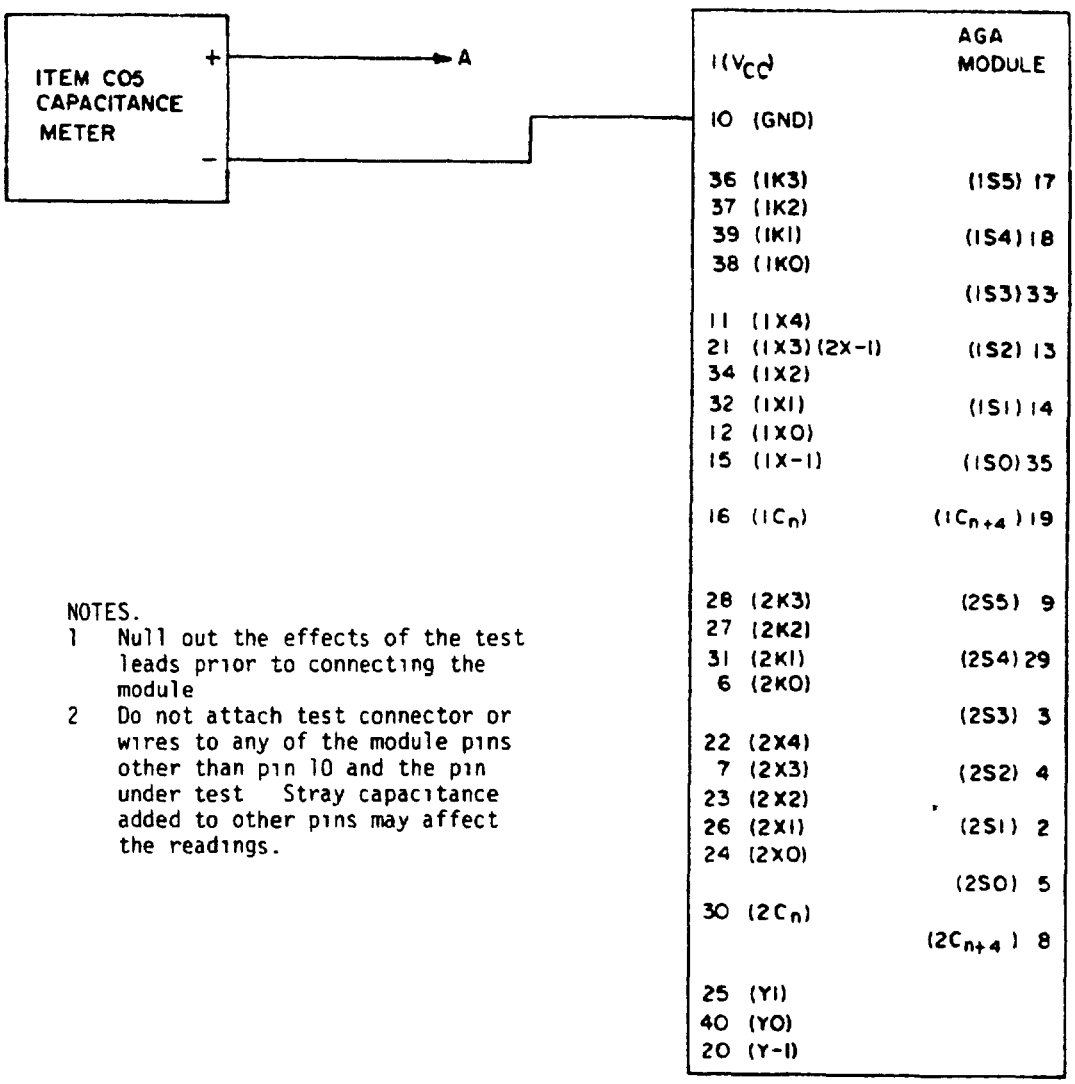
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
S1	1K3	Input capacitance	36	2	15	pF	17
S2	1K2	"	37	"	"	"	"
S3	1K1	"	39	"	"	"	"
S4	1K0	"	38	"	"	"	"
S5	1X4	"	11	"	"	"	"
S6	1X2	"	34	"	"	"	"
S7	1X1	"	32	"	"	"	"
S8	1X0	"	12	"	"	"	"
S9	1X-1	"	15	"	"	"	"
S10	$1C_n$	"	16	"	"		"
S11	2K3	"	28	"	"	"	"
S12	2K2	"	27	"	"	"	"
S13	2K1	"	31	"	"	"	"
S14	2K0	"	6	"	"	"	"
S15	2X4	"	22	"	"	"	"
S16	2X3	"	7	"	"	"	"
S17	2X2	"	23	"	"	"	"
S18	2X1	"	26	"	"	"	"
S19	2X0	"	24	"	"	"	"
S20	$2C_n$	"	30	"	"	"	"

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TABLE XVIII. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
S21	1X3,2X-1 Input capacitance	21	2	30	pF	17
S22	Y1 "	25	"	30	"	"
S23	Y-1 "	20	"	30	"	"
S24	Y0 "	40	"	60	"	"

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NOTES.
1 Null out the effects of the test leads prior to connecting the module
2 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test Stray capacitance added to other pins may affect the readings.

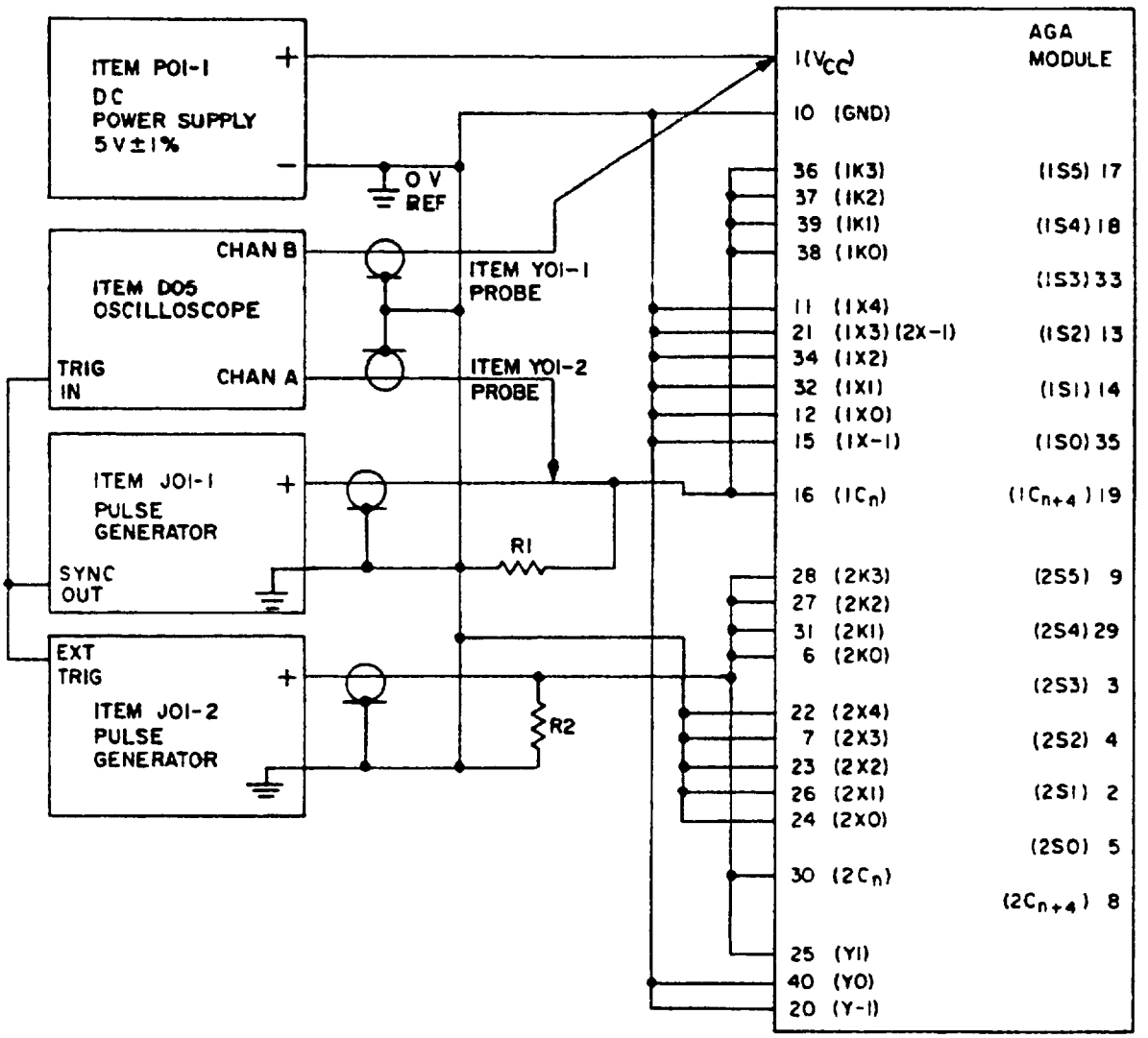
FIGURE 17. Input capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	18,19

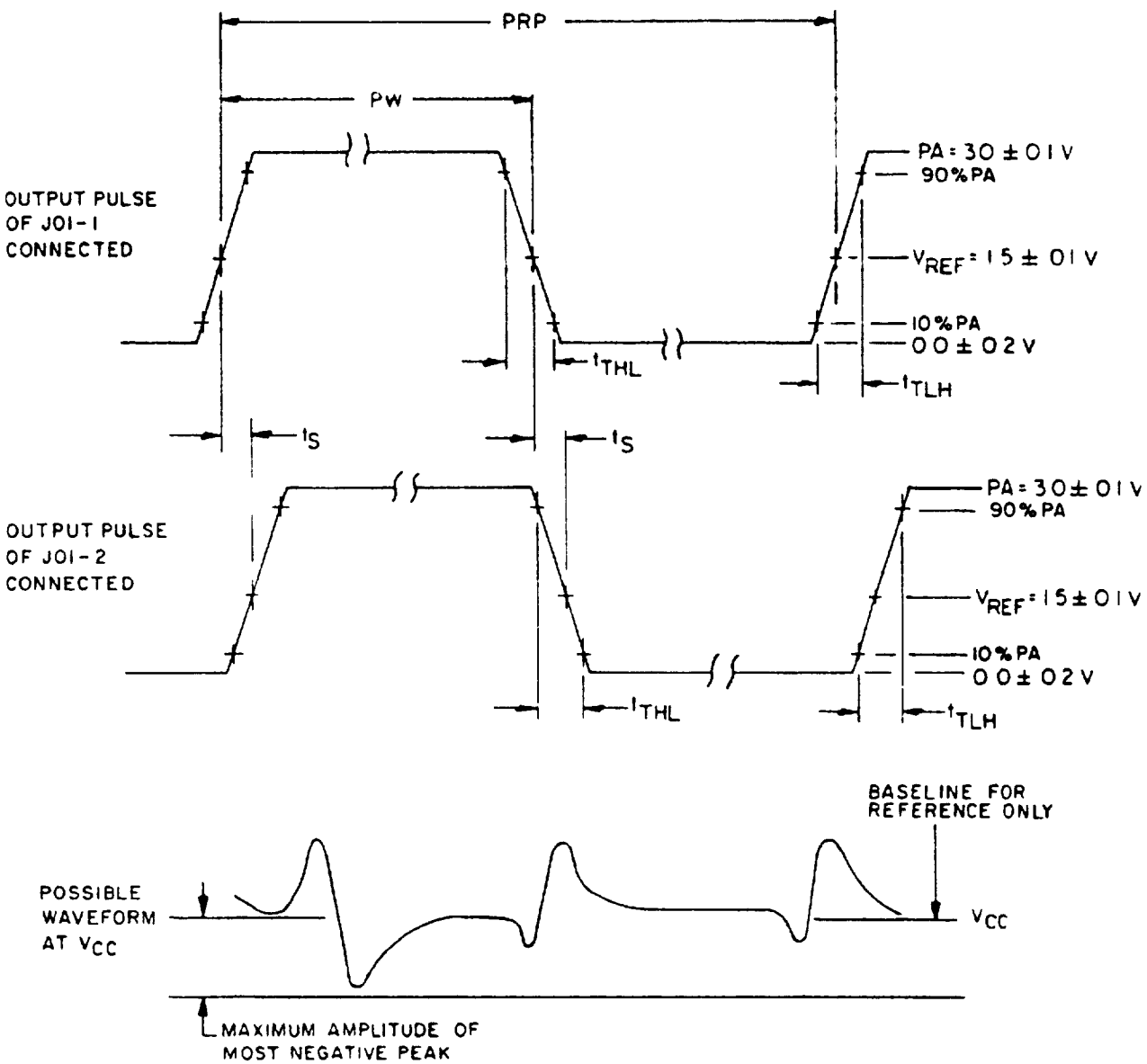
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NOTE R1 = R2 = 51Ω ±5%, 1/2 W

FIGURE 18 Power supply transient amplitude test circuit for table XIX

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NOTE Input pulse parameters (J01-1 and J01-2)
PW = 300 ±30 ns
PRP = 600 ±60 ns
$t_{TLH} = t_{THL}$ = 10 ±2 ns
t_S = 0 ±5 ns.

* FIGURE 19. Power supply transient amplitude test circuit waveforms for table XIX.

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